

Pure- k_T Perturbative TMD Extraction: Formalism Development Notes

k_T Perturbative Extraction Project

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Abstract

These notes give a paper-ready prescription for a purely transverse-momentum-space implementation of Collins/CSS2 TMD factorization. The construction, denoted $\mathcal{K}_{\text{CSS2}}$, uses k_T -space distributions and transverse convolution kernels as the working objects, so that the fitted or modeled representation never has to be a numerical b_T -space object. The local b_T -space Sudakov factor is replaced by a convolution-exponential Green function in k_T space. Perturbative accuracy labels, including $N^3\text{LL}'$, are inherited by ingredient equivalence to the parent CSS2 calculation: the same anomalous dimensions, Collins-Soper kernel, hard factors, TMD matching coefficients, DGLAP ingredients, and fixed-order subtractions must be present. The nonperturbative input is specified as a model-agnostic module, with neural networks, splines, analytic ansatzes, lattice-informed inputs, or other parameterizations treated as interchangeable choices. This version closes the formalism note for use as the basis of a first paper; an actual numerical $N^3\text{LL}'$ extraction remains a separate implementation task requiring staged and validated physical three-loop coefficient payloads.

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1 Status and purpose

This note is the completed formalism blueprint for the first paper. Its purpose is not to report a fitted phenomenological extraction, but to define the prescription that such an extraction must follow if it is to remain in k_T space while carrying the same perturbative accuracy labels as the corresponding Collins/CSS2 calculation. The note therefore fixes:

1. the parent scheme, $\mathcal{K}_{\text{CSS2}}$, and its relation to Collins/CSS2;
2. the k_T -space distribution basis and convolution algebra;
3. the perturbative evolution, matching, and fixed-order expansion checks needed for $N^k\text{LL}$ and $N^k\text{LL}'$ claims;
4. a process-level W -term template that is not restricted to fixed-target Drell-Yan;
5. a model-agnostic nonperturbative module interface;
6. regulated normalization and transverse-moment definitions;
7. the $W + Y$ full-spectrum matching prescription and a perturbative uncertainty decomposition;
8. the claim certificate separating a formal $N^3\text{LL}'$ prescription from a completed numerical $N^3\text{LL}'$ extraction.

The starting phenomenological anchor is the momentum-space neural-network extraction of Fernando and Keller [35]. Their method remains in k_T space, learns a transverse structure kernel from E288/E605 Drell-Yan data, and reconstructs a normalized profile $s(x, k; Q)$ with a differentiable quadrature layer. For the present project, that work is best viewed as an inverse-convolution and nonparametric-extraction prototype. It is not, by itself, an $N^k\text{LL}$ TMD extraction because it does not impose an explicit Sudakov kernel, fixed-order matching, or Y term.

Working decision. The project target is not “a neural network instead of resummation.” The target is a renormalized k_T -space TMD framework whose perturbative content is equivalent to a chosen $N^k\text{LL}$ or $N^k\text{LL}'$ Collins/CSS calculation. The neural network supplies only nonperturbative boundary information.

2 Prior art and project gap

Before deriving new kernels, we separate what is already established from what is specific to this project. The short conclusion is:

Direct transverse-momentum-space resummation exists. High-order perturbative TMD ingredients exist. Neural-network TMD fits exist. Direct- k_T inverse reconstruction exists. The missing object is a universal TMDPDF extraction framework that keeps the fitted representation in k_T space while implementing Collins/CSS-equivalent $N^k\text{LL}$ or $N^k\text{LL}'$ evolution, matching, and $W + Y$ consistency.

2.1 The standard baseline: Collins/CSS and modern high-accuracy TMD fits

The dominant precision framework for small- q_T Drell-Yan is the Collins-Soper-Sterman logic: factorize at low transverse momentum, Fourier transform to b_T space, resum large logarithms through Collins-Soper and RG evolution, and match back to fixed-order collinear factorization outside the strict TMD region [1, 2, 3]. The recent TMD literature provides a broad review of the associated operator definitions, process dependence, evolution, and phenomenological status [4].

This baseline has already reached very high perturbative accuracy in phenomenological extractions. Examples include unpolarized Drell-Yan fits at $N^4\text{LL}$ accuracy that extract TMDPDFs and the Collins-Soper kernel [25], and combined Drell-Yan plus SIDIS analyses at $N^4\text{LL}$ that also include TMD fragmentation functions [26]. Neural-network parametrizations of nonperturbative TMD components have also already been demonstrated at high perturbative accuracy in the standard impact-parameter-space framework [27].

2.2 Direct q_T -space and momentum-space resummation

The idea of resumming directly in transverse momentum is not new. Ellis and Veseli constructed a q_T -space resummation for electroweak boson transverse-momentum spectra and showed agreement with the leading, next-to-leading, and next-to-next-to-leading towers of the usual b -space result [5]. Frixione, Nason, and Ridolfi analyzed analytic issues in both b -space and q_T -space resummation and obtained a closed next-to-leading-logarithmic q_T -space expression, while identifying limitations of its analytic structure [6]. Kulesza and Stirling further explored explicit transverse-momentum-space Sudakov resummation for electroweak boson production and its matching behavior [7, 8].

Modern direct-space approaches made this line much more powerful. Monni, Re, and Torrielli formulated a momentum-space resummation for Higgs transverse momentum at NNLL+NNLO [9]. Bizon, Monni, Re, Rottoli, and Torrielli then generalized momentum-space resummation for transverse observables up to $N^3\text{LL}+\text{NNLO}$, with a formula that reduces to the standard impact-parameter-space result in known cases [10]. Ebert and Tackmann developed a related distribution-space RG method in which the singular plus distributions in q_T are treated directly as the objects of evolution [11]. More recently, Simonelli proposed an analytic transverse-momentum-space continuation of TMDPDFs and TMD-factorized cross sections to NLL accuracy [12].

2.3 Momentum-space TMD ingredients and high-order perturbative inputs

There is also substantial prior work on perturbative TMD ingredients that can be used as input to a direct- k_T construction. In the rapidity-renormalization-group formulation, rapidity-renormalized TMD soft and beam functions were computed at two loops, providing NNLL' ingredients for transverse-momentum-differential predictions [16]. TMDPDF matching coefficients are now known to very high order: Ebert, Mistlberger, and Vita computed quark and gluon TMDPDFs at $N^3\text{LO}$ [17], and Luo, Yang, Zhu, and Zhu computed unpolarized quark and gluon TMDPDF and TMDFF matching coefficients through $N^3\text{LO}$ [19]. The rapidity anomalous dimension, equivalently the perturbative Collins-Soper kernel in common schemes, is also known at very high order [20, 21].

For our purposes, the key lesson from this literature is that the perturbative ingredients should not be refit by the neural network. They should enter as hard, evolution, matching, and subtraction kernels. The nonperturbative model should only cover the long-distance or small- k_T boundary information left unresolved by perturbation theory.

2.4 Large- k_T consistency, integral constraints, and hadron-structure-oriented fits

A central issue for a pure- k_T extraction is the transition between the nonperturbative core and the perturbative large- k_T tail. Gonzalez-Hernandez, Rainaldi, and Rogers stressed that TMD parametrizations must be made consistent with their large-transverse-momentum behavior in collinear factorization [23]. The hadron-structure-oriented approach implements related constraints in practical Drell-Yan and Z^0 phenomenology, emphasizing preservation of the TMD-PDF integral relationship and controlled comparisons among nonperturbative transverse structures [24]. Ebert, Michel, Stewart, and Sun likewise showed that momentum-space TMD integral functionals provide a systematic way to disentangle long and short distances, replacing naive unregulated integral constraints by regulated, perturbatively controlled objects [22].

2.5 Neighboring momentum-space phenomenology

The Parton Branching approach is an important neighboring framework because it evolves TMD parton densities in momentum space and has been used to describe Drell-Yan transverse momentum spectra across small and large k_T regions [29]. However, PB TMDs and CSS TMDs are not the same formal objects. A direct comparison paper emphasizes that PB TMDs evolve with an explicit single scale in momentum space, whereas CSS TMDs have coupled renormalization and rapidity-scale evolution [28]. For this project, PB is prior art and a useful comparison target, but not a replacement for a Collins-equivalent double-scale k_T -space TMD factorization framework.

Fernando and Keller provide the closest direct predecessor on the extraction side: they perform a physics-informed neural inverse reconstruction directly in momentum space, using fixed-target Drell-Yan data and a normalized transverse profile whose auto-convolution reproduces the measured spectra [35]. Their construction is precisely why the present project starts from a direct- k_T inverse problem. The perturbative upgrade is to replace their tree-level/profile-only forward model by a distribution-valued $N^k\text{LL}_{k_T}$ forward operator.

2.6 Gap statement

The prior art implies a useful division of labor, summarized in Table 1. The project should not claim to invent momentum-space resummation, high-order TMD matching, or neural-network TMD fitting. The defensible claim is the synthesis of these ingredients into a TMD-level extraction architecture.

Table 1: Prior art and the remaining project gap.

Prior-art line	What already exists	What remains for this project
CSS/Collins TMD factorization	Double-scale TMD evolution, small- q_T factorization, OPE matching, and $W + Y$ logic in a b_T -space representation [2, 3].	Represent the same perturbative content by k_T -space convolution kernels and fit the boundary object directly in k_T .

Prior-art line	What already exists	What remains for this project
Direct q_T resummation	Cross-section-level resummation directly in q_T or momentum space, including modern high-accuracy direct-space implementations [5, 9, 10].	Promote the direct-space logic from color-singlet spectra to universal TMDPDF evolution and extraction.
Distribution-space RG	Plus-distribution evolution kernels and distributional scale setting for transverse-momentum spectra [11].	Turn the distribution algebra into a practical TMDPDF Green's function $\mathcal{U}_i(\mathbf{k}; \mu_f, \zeta_f; \mu_i, \zeta_i)$.
High-order TMD ingredients	Soft/beam functions, matching coefficients, and rapidity anomalous dimensions through high perturbative orders [16, 17, 19, 20].	Convert and organize these inputs consistently in the chosen k_T -space scheme.
Modern TMD fits	$N^3\text{LL}$ and $N^4\text{LL}$ phenomenological extractions, including neural nonperturbative parametrizations [25, 26, 27].	Keep the fitted representation in k_T and make the neural model a boundary core rather than the full TMD.
Direct- k_T inverse reconstruction	Fernando-Keller learn a normalized k_T profile from fixed-target Drell-Yan data without a b_T transform [35].	Add perturbative evolution, large- k_T matching, regulated integral constraints, and Y -term consistency.
Parton Branching	Momentum-space TMD evolution and Drell-Yan phenomenology with an explicit single evolution scale [29, 28].	Build a Collins/CSS-equivalent two-scale (μ, ζ) momentum-space TMD framework rather than a PB-to-CSS conversion.

The target object is therefore the simultaneous implementation of

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} F_{i/h}(x, \mathbf{k}; \mu, \zeta) = -[\mathcal{D}_i(\mu) \otimes_T F_{i/h}(x; \mu, \zeta)](\mathbf{k}), \quad (1)$$

$$F_{i/h}(x, \mathbf{k}; \mu_0, \zeta_0) = \sum_j \mathcal{C}_{i \leftarrow j}(\mathbf{k}; \mu_0, \zeta_0) \otimes_x f_{j/h}(\mu_0) + F_{i/h}^{\text{core}}(x, \mathbf{k}; \theta) - F_{i/h}^{\text{sub}}(x, \mathbf{k}; \theta), \quad (2)$$

$$\frac{d\sigma}{dQ^2 dy d^2\mathbf{q}} = \sigma_0 \left[W_{k_T}^{N^k\text{LL}}(Q, y, \mathbf{q}) + Y_{k_T}(Q, y, \mathbf{q}) \right] + \mathcal{O}(q_T^2/Q^2), \quad (3)$$

with all singular kernels treated as transverse-momentum distributions and all nonperturbative information supplied by a fitted boundary core.

Working decision. Claim discipline: we do not claim the first momentum-space resummation, the first momentum-space TMD calculation, or the first neural-network TMD fit. We claim a TMD-level, Collins-equivalent, pure- k_T extraction framework that synthesizes those lines of work.

3 Core thesis

The Collins/CSS advantage is not fundamentally the coordinate b_T itself [1, 2, 3]. The advantage is that Fourier transformation converts transverse convolutions into products, making Collins-Soper and renormalization-group evolution local in transverse separation. A pure k_T formulation must keep the same perturbative content while accepting a different mathematical structure:

$$\text{local multiplication in } b_T \iff \text{convolution operators in } k_T. \quad (4)$$

The working thesis is therefore

$$N^k \text{LL}_{k_T} \equiv N^k \text{LL}_{b_T} \text{ perturbative content represented by } k_T\text{-space distributions.}$$

That means the same hard functions, anomalous dimensions, matching coefficients, and $W + Y$ subtractions as the corresponding Collins/CSS calculation.

4 Paper-ready working choices

The following choices define the paper-level prescription. They are conservative by design: we first represent Collins/CSS2 directly in k_T space and require explicit conversion factors before introducing any child scheme. These choices are broad enough for a formal paper and narrow enough for a later implementation repository to test.

Table 2: Paper-ready working choices.

Issue	Working choice
Reference scheme	Use Collins/CSS2 as the parent scheme. Define the initial pure- k_T implementation, denoted $\mathcal{K}_{\text{CSS2}}$, as the momentum-space representation of the Collins/CSS2 TMDs, hard factors, Collins-Soper kernel, and matching coefficients. This follows the scheme-translation logic of Collins and Rogers [36].
Custom scheme	Do not introduce a nontrivial finite custom conversion at the first step. First build $\mathcal{K}_{\text{CSS2}}$ and verify exact equivalence to CSS2 at fixed perturbative order. A more adapted child scheme $\mathcal{K}$ may be introduced later only with explicit finite kernels $\mathcal{Z}^{\mathcal{K} \leftarrow \mathcal{K}_{\text{CSS2}}}$ and inverse kernels.
Distribution basis	Keep both the conventional two-dimensional plus distributions $\mathcal{L}_n$ and the analytic generator $\mathcal{G}_\eta$. Use $\mathcal{L}_n$ for comparison with singular coefficients and bin integrals; use $\mathcal{G}_\eta$ for convolution algebra and exponentiation.
Perturbative tail	Define the universal TMD large- k_T tail from TMD matching coefficients, $\mathcal{C}_{i \leftarrow j}^{\mathcal{K}_{\text{CSS2}}} \otimes_x f_j$. Do not use a cross-section generator to define the TMD tail.

Issue	Working choice
Large- q_T matching	Define the cross-section tail through the usual Y term, $Y = \sigma_{\text{FO}} - W_{\text{FO exp.}}$. Use DYTurbo, CuTe-MCFM/MCFM, or equivalent fixed-order/resummed tools as validation and benchmarking references, not as definitions of the universal TMDPDF [37, 38, 39].
Normalization	Use regulated cumulants and weighted moments. Do not impose an unregulated full- k_T integral as a fundamental identity. The same definitions must work from fixed-target kinematics to collider kinematics.
Uncertainty	In the formalism stage, track perturbative uncertainties only: scale/path choices, matching truncation, scheme-conversion truncation, and distributional numerical errors. Leave neural/TMD model, experimental, and replica uncertainties for the later fitting stage.
Flavor	Use a full flavor-vector formalism from day one. Fixed-target Drell-Yan can be the first data application, but the formalism must also accommodate collider Drell-Yan, SIDIS, e^+e^- , and gluon-sensitive color-singlet channels.

Working decision. The first concrete scheme is $\mathcal{K}_{\text{CSS2}}$: a pure- k_T representation of Collins/CSS2. This is deliberately conservative. It lets us prove equivalence to the standard formalism before making any finite, numerically adapted scheme redefinitions.

5 Conventions

Transverse vectors are bold: $\mathbf{k}, \mathbf{q}, \mathbf{p}, \mathbf{b}$. Their magnitudes are denoted by $k_T = |\mathbf{k}|$, $q_T = |\mathbf{q}|$, etc. The transverse convolution is

$$(A \otimes_T B)(\mathbf{q}) \equiv \int d^2\mathbf{k} A(\mathbf{k}) B(\mathbf{q} - \mathbf{k}). \quad (5)$$

The longitudinal convolution is

$$(C \otimes_x f)(x) \equiv \int_x^1 \frac{dz}{z} C(z) f\left(\frac{x}{z}\right). \quad (6)$$

For derivations and cross-checks we may use the Fourier pair

$$\tilde{F}(x, \mathbf{b}; \mu, \zeta) = \int d^2\mathbf{k} e^{i\mathbf{b} \cdot \mathbf{k}} F(x, \mathbf{k}; \mu, \zeta), \quad (7)$$

$$F(x, \mathbf{k}; \mu, \zeta) = \int \frac{d^2\mathbf{b}}{(2\pi)^2} e^{-i\mathbf{b} \cdot \mathbf{k}} \tilde{F}(x, \mathbf{b}; \mu, \zeta). \quad (8)$$

The fitted representation and production code should remain in k_T space. Fourier-space expressions are allowed as derivational checks and as sources of known perturbative coefficients.

We use Drell-Yan rapidity scales satisfying

$$\zeta_A \zeta_B = Q^4. \quad (9)$$

The symmetric choice is $\zeta_A = \zeta_B = Q^2$, but the formalism should support arbitrary rapidity-scale profiles for uncertainty estimates and path-independence checks.

6 Reference scheme: KCSS2 to Collins/CSS2

The initial scheme is not a new operator definition. It is the Collins/CSS2 TMD scheme represented directly in transverse-momentum space. Collins/CSS2 denotes the modern Collins-style TMD factorization convention discussed in Collins' book and in the scheme-conversion analysis of Collins and Rogers [3, 36]. The older CSS1 coefficient notation is useful historically, but our parent scheme is CSS2.

Define the inverse transverse Fourier map by

$$\mathcal{F}_{b \rightarrow k}^{-1}[\tilde{A}](\mathbf{k}) \equiv \int \frac{d^2 \mathbf{b}}{(2\pi)^2} e^{-i\mathbf{b} \cdot \mathbf{k}} \tilde{A}(\mathbf{b}), \quad \mathcal{F}_{k \rightarrow b}[A](\mathbf{b}) \equiv \int d^2 \mathbf{k} e^{i\mathbf{b} \cdot \mathbf{k}} A(\mathbf{k}). \quad (10)$$

The defining relation for the reference k_T scheme is

$$F_{i/h}^{[\mathcal{K}_{\text{CSS2}}]}(x, \mathbf{k}; \mu, \zeta) \equiv \mathcal{F}_{b \rightarrow k}^{-1} \left[\tilde{F}_{i/h}^{[\text{CSS2}]}(x, \mathbf{b}; \mu, \zeta) \right](\mathbf{k}), \quad (11)$$

where the equality is distributional. The same mapping defines the perturbative kernels:

$$\mathcal{D}_i^{[\mathcal{K}_{\text{CSS2}}]}(\mathbf{k}; \mu) \equiv \mathcal{F}_{b \rightarrow k}^{-1} \left[D_i^{[\text{CSS2}]}(\mathbf{b}; \mu) \right](\mathbf{k}), \quad (12)$$

$$\mathcal{C}_{i \leftarrow j}^{[\mathcal{K}_{\text{CSS2}}]}(z, \mathbf{k}; \mu, \zeta) \equiv \mathcal{F}_{b \rightarrow k}^{-1} \left[\tilde{C}_{i \leftarrow j}^{[\text{CSS2}]}(z, \mathbf{b}; \mu, \zeta) \right](\mathbf{k}). \quad (13)$$

Here $D_i^{[\text{CSS2}]}$ is defined with the sign convention

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} \tilde{F}_{i/h}^{[\text{CSS2}]}(x, \mathbf{b}; \mu, \zeta) = -D_i^{[\text{CSS2}]}(\mathbf{b}; \mu) \tilde{F}_{i/h}^{[\text{CSS2}]}(x, \mathbf{b}; \mu, \zeta). \quad (14)$$

If a reference uses Collins' original $\tilde{K}$ notation with the opposite sign, then $D_i^{[\text{CSS2}]} = -\tilde{K}_i$ in the present convention.

With these definitions, the usual CSS2 W term

$$W_{AB}^{[\text{CSS2}]}(Q, y, \mathbf{q}) = \sum_{ij} H_{ij}^{[\text{CSS2}]}(Q, \mu) \int \frac{d^2 \mathbf{b}}{(2\pi)^2} e^{-i\mathbf{b} \cdot \mathbf{q}} \tilde{F}_{i/A}^{[\text{CSS2}]}(x_A, \mathbf{b}; \mu, \zeta_A) \tilde{F}_{j/B}^{[\text{CSS2}]}(x_B, \mathbf{b}; \mu, \zeta_B) \quad (15)$$

becomes exactly

$$W_{AB}^{[\mathcal{K}_{\text{CSS2}}]}(Q, y, \mathbf{q}) = \sum_{ij} H_{ij}^{[\text{CSS2}]}(Q, \mu) \left[F_{i/A}^{[\mathcal{K}_{\text{CSS2}}]}(x_A; \mu, \zeta_A) \otimes_T F_{j/B}^{[\mathcal{K}_{\text{CSS2}}]}(x_B; \mu, \zeta_B) \right](\mathbf{q}). \quad (16)$$

Thus the first k_T implementation uses the same hard function as CSS2 and simply represents the two TMD factors and their product by transverse convolutions.

6.1 Optional child schemes and finite conversion kernels

A later numerically adapted scheme may be defined by finite, process-independent kernels,

$$F_{i/h}^{[\mathcal{K}]}(x, \mathbf{k}; \mu, \zeta) = \sum_j \left[\mathcal{Z}_{i \leftarrow j}^{\mathcal{K} \leftarrow \mathcal{K}_{\text{CSS2}}} \otimes_x \otimes_T F_{j/h}^{[\mathcal{K}_{\text{CSS2}}]} \right](x, \mathbf{k}; \mu, \zeta), \quad (17)$$

with

$$\mathcal{Z}_{i \leftarrow j}^{\mathcal{K} \leftarrow \mathcal{K}_{\text{CSS2}}}(z, \mathbf{k}) = \delta_{ij} \delta(1-z) \delta^{(2)}(\mathbf{k}) + \mathcal{O}(\alpha_s). \quad (18)$$

The inverse relation is

$$F_{i/h}^{[\mathcal{K}_{\text{CSS2}}]} = \sum_j \left[\left(\mathcal{Z}^{\mathcal{K} \leftarrow \mathcal{K}_{\text{CSS2}}} \right)_{i \leftarrow j}^{-1} \otimes_x \otimes_T F_{j/h}^{[\mathcal{K}]} \right]. \quad (19)$$

If $\mathcal{Z}$ has nontrivial transverse-momentum dependence, the physical W term written in terms of $F^{[\mathcal{K}]}$ contains these inverse kernels explicitly:

$$W_{AB} = \sum_{ij} H_{ij}^{[\text{CSS2}]} \left[\left(\mathcal{Z}^{-1} \otimes_x \otimes_T F_A^{[\mathcal{K}]} \right)_i \otimes_T \left(\mathcal{Z}^{-1} \otimes_x \otimes_T F_B^{[\mathcal{K}]} \right)_j \right](\mathbf{q}). \quad (20)$$

Therefore a nonlocal transverse conversion kernel cannot simply be hidden inside a b -independent hard function. This is why the first implementation sets $\mathcal{Z} = \mathbf{1}$ and works in $\mathcal{K}_{\text{CSS2}}$. After the one-loop and two-loop equivalence tests are passed, we can decide whether a nontrivial child scheme is worth the extra bookkeeping.

Working decision. Separate three notions that are often conflated: the TMD operator scheme, the transverse distribution basis, and the numerical implementation. The first implementation keeps the Collins/CSS2 operator scheme, derives the one-loop k_T -space matching and Collins-Soper kernels, validates the bin and flavor-vector checks, and now adds the first one-loop evolution Green function.

6.2 First conversion targets

The $\mathcal{K}_{\text{CSS2}} \leftrightarrow \text{CSS2}$ bridge reduces the first derivations to a finite list:

1. validate the mapped one-loop CSS2 quark TMD matching coefficient $\tilde{C}_{q \leftarrow q}^{[\text{CSS2}]}(z, b)$ and $\tilde{C}_{q \leftarrow g}^{[\text{CSS2}]}(z, b)$ into the $\mathcal{L}_n$ and $\mathcal{G}_\eta$ bases;
2. map the perturbative Collins-Soper kernel $D_q^{[\text{CSS2}]}(b; \mu)$ into $\mathcal{D}_q^{[\mathcal{K}_{\text{CSS2}}]}(k; \mu)$;
3. verify that Eq. (16), expanded through $\mathcal{O}(\alpha_s)$, reproduces the known Drell-Yan small- q_T singular spectrum;
4. add the Y term only after the singular W operator passes the CSS2 equivalence tests.

7 Momentum-space TMD factorization for Drell-Yan

For a color-singlet Drell-Yan final state with invariant mass Q , rapidity y , and small transverse momentum $q_T \ll Q$, define

$$x_A = \frac{Q}{\sqrt{s}} e^y, \quad x_B = \frac{Q}{\sqrt{s}} e^{-y}. \quad (21)$$

The singular or W term is written directly in transverse momentum as

$$W_{AB}(Q, y, \mathbf{q}; \mu, \zeta_A, \zeta_B) = \sum_{ij} H_{ij}(Q, \mu) \int d^2 \mathbf{k}_A d^2 \mathbf{k}_B \delta^{(2)}(\mathbf{q} - \mathbf{k}_A - \mathbf{k}_B) \\ \times F_{i/A}^{\text{DY}}(x_A, \mathbf{k}_A; \mu, \zeta_A) F_{j/B}^{\text{DY}}(x_B, \mathbf{k}_B; \mu, \zeta_B). \quad (22)$$

Equivalently,

$$W_{AB} = \sum_{ij} H_{ij} [F_{i/A}^{\text{DY}}(x_A; \mu, \zeta_A) \otimes_T F_{j/B}^{\text{DY}}(x_B; \mu, \zeta_B)](\mathbf{q}). \quad (23)$$

The matched cross section is organized as

$$\frac{d\sigma}{dQ^2 dy d^2\mathbf{q}} = \sigma_0(Q, s, y) [W_{AB}^{\text{resum}}(Q, y, \mathbf{q}) + Y_{AB}(Q, y, \mathbf{q})] + \mathcal{O}\left(\frac{q_T^2}{Q^2}\right). \quad (24)$$

The Y term is defined by the standard subtraction identity

$$Y_{AB} = \left[\frac{1}{\sigma_0} \frac{d\sigma}{dQ^2 dy d^2\mathbf{q}} \right]_{\text{FO}} - [W_{AB}^{\text{resum}}]_{\text{expanded to same FO}}. \quad (25)$$

This definition is independent of whether W is represented in b_T or k_T space.

Working decision. Fixed-target-only studies can initially use the W term in a restricted small- q_T region, matching the scope of Fernando-Keller. A globally competitive formalism must include Eq. (25).

8 Generic process-level W operator

Drell-Yan is the first validation channel, but the formalism is not DY-specific. The universal object is a process-level singular operator built from perturbatively evolved TMD legs and a process-dependent hard function. For any color-singlet process whose leading-power small- q_T factorization contains two transverse-momentum-carrying legs, we write schematically

$$W_F^{\mathcal{K}_{\text{CSS2}}}(\mathbf{q}; Q, \{x\}, \mu) = \sum_{ab} H_{ab \rightarrow F}(Q, \mu) \left[\mathbb{F}_{a/A}^{\mathcal{K}_{\text{CSS2}}}(x_A; Q) \otimes_T \mathbb{F}_{b/B}^{\mathcal{K}_{\text{CSS2}}}(x_B; Q) \right](\mathbf{q}), \quad (26)$$

where F denotes the measured color-singlet final state and where each leg $\mathbb{F}$ is understood to include the chosen perturbative boundary, evolution from the boundary point to the hard point, and any admissible nonperturbative module. The same algebra applies when a leg is a TMD fragmentation function rather than a TMDPDF.

The main specializations are:

$$\text{Drell-Yan:} \quad W_{\text{DY}} = \sum_q e_q^2 H_{q\bar{q}}^{\text{DY}} \left[F_{q/A} \otimes_T F_{\bar{q}/B} + F_{\bar{q}/A} \otimes_T F_{q/B} \right], \quad (27)$$

$$\text{SIDIS:} \quad W_{\text{SIDIS}} = \sum_{ij} H_{ij}^{\text{SIDIS}} \left[F_{i/A} \otimes_T D_{h/j} \right], \quad (28)$$

$$e^+e^- \rightarrow h_1 h_2 X : \quad W_{e^+e^-} = \sum_{ij} H_{ij}^{e^+e^-} \left[D_{h_1/i} \otimes_T D_{h_2/j} \right], \quad (29)$$

$$\text{gluon color-singlet:} \quad W_{gg \rightarrow F} = H_{gg \rightarrow F} \left[F_{g/A} \otimes_T F_{g/B} \right] + \text{quark-channel terms as required.} \quad (30)$$

The hard functions and Wilson-line directions are process dependent; the distribution basis, transverse convolution algebra, and accuracy bookkeeping are not. A first DY implementation is therefore a validation channel for the generic $\mathcal{K}_{\text{CSS2}}$ operator, not a restriction of the framework.

For later phenomenology, it is useful to collect partons into a flavor vector,

$$\mathbf{F}_h = (F_{g/h}, F_{u/h}, F_{\bar{u}/h}, F_{d/h}, F_{\bar{d}/h}, \dots)^T, \quad (31)$$

and to regard $\mathcal{C}_{i \leftarrow j}$, DGLAP evolution, and scheme conversions as matrices in this vector space. Flavor simplifications may be introduced in a data fit, but not in the definition of the formalism.

9 Evolution as convolutional dynamics

In a conventional b_T representation, Collins-Soper evolution is local in b_T . In momentum space, the same physics becomes nonlocal in transverse momentum:

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} F_i(x, \mathbf{k}; \mu, \zeta) = -[\mathcal{D}_i(\mu) \otimes_T F_i(x; \mu, \zeta)](\mathbf{k}), \quad (32)$$

where $\mathcal{D}_i(\mathbf{k}; \mu)$ is the momentum-space Collins-Soper kernel. The sign and normalization of $\mathcal{D}_i$ define a convention; all hard, soft, and matching coefficients must be converted consistently.

The renormalization-scale equation is local in transverse momentum,

$$\frac{\partial}{\partial \ln \mu} F_i(x, \mathbf{k}; \mu, \zeta) = \gamma_F^i(\mu, \zeta) F_i(x, \mathbf{k}; \mu, \zeta), \quad (33)$$

while the Collins-Soper kernel satisfies, in the Collins/CSS convention,

$$\frac{d}{d \ln \mu} \mathcal{D}_i(\mathbf{k}; \mu) = \gamma_K^i(\alpha_s(\mu)) \delta^{(2)}(\mathbf{k}) = 2\Gamma_{\text{cusp}}^i(\alpha_s(\mu)) \delta^{(2)}(\mathbf{k}), \quad (34)$$

where γ_K^i is Collins' kernel anomalous dimension and Γ_{cusp}^i is the lightlike cusp anomalous dimension. This factor-of-two bookkeeping is important when comparing CSS notation to SCET notation.

At fixed μ , the solution of Eq. (32) is a convolution exponential,

$$F_i(x, \mathbf{k}; \mu, \zeta_f) = \left[\text{Exp}_{\otimes_T} \left\{ -\ln \left(\sqrt{\zeta_f / \zeta_i} \right) \mathcal{D}_i(\mu) \right\} \otimes_T F_i(x; \mu, \zeta_i) \right](\mathbf{k}), \quad (35)$$

where

$$\text{Exp}_{\otimes_T} \{A\} = \delta^{(2)} + A + \frac{1}{2!} A \otimes_T A + \frac{1}{3!} A \otimes_T A \otimes_T A + \dots \quad (36)$$

Working decision. A central implementation object is the k_T -space evolution Green's function $\mathcal{U}_i(\mathbf{k}; \mu_f, \zeta_f; \mu_i, \zeta_i)$ satisfying $F_i(\mu_f, \zeta_f) = \mathcal{U}_i \otimes_T F_i(\mu_i, \zeta_i)$. The Green's function is a radial distribution, not an ordinary pointwise function.

10 Distribution basis for pure- k_T resummation

Perturbative transverse-momentum spectra contain singular distributions of the form [11]

$$\left[\frac{1}{k_T^2} \ln^n \left(\frac{\mu^2}{k_T^2} \right) \right]_+ \quad \text{and} \quad \delta^{(2)}(\mathbf{k}). \quad (37)$$

A convenient two-dimensional basis is

$$\mathcal{L}_n(\mathbf{k}, \mu) = \frac{1}{\pi \mu^2} \left[\frac{\mu^2}{k_T^2} \ln^n \left(\frac{\mu^2}{k_T^2} \right) \right]_+, \quad n \geq 0. \quad (38)$$

The precise plus prescription must be tied to binned observables or smooth test functions; pointwise values at $k_T = 0$ are not physical.

For algebraic manipulations it may be better to define a generator whose Fourier transform is exactly a power of the standard b_T logarithm. Let

$$L_b \equiv \ln \left(\frac{\mu^2 b_T^2 e^{2\gamma_E}}{4} \right) \quad (39)$$

and define $\mathcal{G}_\eta$ distributionally by

$$\int d^2\mathbf{k} e^{i\mathbf{b}\cdot\mathbf{k}} \mathcal{G}_\eta(\mathbf{k}; \mu) = e^{-\eta L_b}. \quad (40)$$

Then

$$\mathcal{G}_\eta \otimes_T \mathcal{G}_{\eta'} = \mathcal{G}_{\eta+\eta'}. \quad (41)$$

A candidate real-space representation, valid first for a regulated range of η and then by distributional continuation, is

$$\mathcal{G}_\eta(\mathbf{k}; \mu) = \frac{1}{\pi\mu^2} \left(\frac{\mu^2}{k_T^2} \right)^{1-\eta} \frac{e^{-2\gamma_E\eta} \Gamma(1-\eta)}{\Gamma(\eta)}. \quad (42)$$

This convention packages many delta-function constants into the basis. It should be mapped explicitly onto Eq. (38) before comparison with published singular coefficients.

Expanding Eq. (40) gives a distribution basis with simple convolution algebra:

$$\mathfrak{L}_n(\mathbf{k}; \mu) \equiv (-1)^{n+1} \left. \frac{\partial^{n+1}}{\partial \eta^{n+1}} \mathcal{G}_\eta(\mathbf{k}; \mu) \right|_{\eta=0}, \quad \int d^2\mathbf{k} e^{i\mathbf{b}\cdot\mathbf{k}} \mathfrak{L}_n(\mathbf{k}; \mu) = L_b^{n+1}. \quad (43)$$

The expansion of Eq. (42) in the conventional basis is

$$\mathcal{G}_\eta = h(\eta) \left[\delta^{(2)}(\mathbf{k}) + \eta \mathcal{L}_0(\mathbf{k}, \mu) - \eta^2 \mathcal{L}_1(\mathbf{k}, \mu) + \frac{\eta^3}{2} \mathcal{L}_2(\mathbf{k}, \mu) + \mathcal{O}(\eta^4) \right], \quad (44)$$

where

$$h(\eta) \equiv e^{-2\gamma_E\eta} \frac{\Gamma(1-\eta)}{\Gamma(1+\eta)} = 1 + \frac{2\zeta_3}{3} \eta^3 + \mathcal{O}(\eta^5). \quad (45)$$

Thus, for the first few basis elements,

$$\mathfrak{L}_0 = -\mathcal{L}_0, \quad (46)$$

$$\mathfrak{L}_1 = -2\mathcal{L}_1, \quad (47)$$

$$\mathfrak{L}_2 = -3\mathcal{L}_2 - 4\zeta_3 \delta^{(2)}. \quad (48)$$

Equivalently,

$$\mathcal{F}_{k \rightarrow b}[\mathcal{L}_0] = -L_b, \quad (49)$$

$$\mathcal{F}_{k \rightarrow b}[\mathcal{L}_1] = -\frac{1}{2} L_b^2, \quad (50)$$

$$\mathcal{F}_{k \rightarrow b}[\mathcal{L}_2] = -\frac{1}{3} L_b^3 - \frac{4}{3} \zeta_3. \quad (51)$$

The signs follow from our choice to define $\mathcal{L}_n$ with $\ln^n(\mu^2/k_T^2)$ rather than $\ln^n(k_T^2/\mu^2)$. This is the most common source of sign mistakes when comparing formulas across the direct- q_T resummation literature.

11 One-loop KCSS2 kernels

This section performs the first explicit bridge calculation. We start from the CSS2 small- b_T coefficients and Collins-Soper kernel, then represent them as distributions in k_T space. Throughout this section

$$a_s(\mu) \equiv \frac{\alpha_s(\mu)}{4\pi}, \quad b_0 \equiv 2e^{-\gamma_E}, \quad L_b \equiv \ln\left(\frac{\mu^2 b_T^2}{b_0^2}\right), \quad \ell_\zeta \equiv \ln\left(\frac{\zeta}{\mu^2}\right). \quad (52)$$

The leading-order collinear splitting functions are defined in the standard $df/d\ln\mu^2 = (\alpha_s/2\pi)P^{(0)} \otimes f + \dots$ convention:

$$P_{qq}^{(0)}(z) = C_F \left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2}\delta(1-z) \right], \quad (53)$$

$$P_{gg}^{(0)}(z) = T_F \left[z^2 + (1-z)^2 \right]. \quad (54)$$

11.1 CSS2 input at the canonical small-bT point

Collins and Rogers give the one-loop CSS2 matching coefficients at the canonical point [36]

$$\mu_b = \frac{b_0}{b_T}, \quad \zeta_b = \mu_b^2 = \frac{b_0^2}{b_T^2}, \quad L_b = \ell_\zeta = 0. \quad (55)$$

In our notation their Eq. (74) gives

$$\tilde{C}_{q \leftarrow q}^{(1), \text{CSS2}}(z; \mu_b, \zeta_b) = C_F \left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z) \right], \quad (56)$$

$$\tilde{C}_{q \leftarrow g}^{(1), \text{CSS2}}(z; \mu_b, \zeta_b) = 2z(1-z), \quad (57)$$

$$\tilde{C}_{q \leftarrow q'}^{(1), \text{CSS2}} = \tilde{C}_{q \leftarrow q}^{(1), \text{CSS2}} = \tilde{C}_{q \leftarrow \bar{q}'}^{(1), \text{CSS2}} = 0. \quad (58)$$

The coefficient is expanded as

$$\tilde{C}_{i \leftarrow j}^{\text{CSS2}} = \delta_{ij}\delta(1-z) + a_s \tilde{C}_{i \leftarrow j}^{(1), \text{CSS2}} + \mathcal{O}(a_s^2). \quad (59)$$

The $-\pi^2/6$ constant is part of the CSS2 separation between TMD matching and the Drell-Yan hard factor. A different finite TMD scheme would reshuffle this delta-function constant, but not the logarithmic distributional structure derived below.

11.2 Lifting the canonical coefficient to general mu and zeta

The logarithms at general (μ, ζ) are fixed by the CSS2 coefficient evolution equations [36]. To one loop, integrating those equations from the canonical point gives

$$\tilde{C}_{q \leftarrow q}^{(1), \text{CSS2}}(z, b_T; \mu, \zeta) = C_F \left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z) \right] - 2L_b P_{qq}^{(0)}(z) \quad (60)$$

$$+ C_F \delta(1-z) \left[(3 - 2\ell_\zeta)L_b - L_b^2 \right], \quad (61)$$

$$\tilde{C}_{q \leftarrow g}^{(1), \text{CSS2}}(z, b_T; \mu, \zeta) = 2z(1-z) - 2L_b P_{qg}^{(0)}(z), \quad (62)$$

with the same zero entries as Eq. (58). These expressions obey two immediate checks:

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} \tilde{C}_{q \leftarrow q}^{(1), \text{CSS2}} = -4C_F L_b \delta(1-z), \quad (63)$$

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} \tilde{C}_{q \leftarrow g}^{(1), \text{CSS2}} = 0, \quad (64)$$

which is the one-loop CSS equation because

$$\widetilde{K}_q^{(1)}(b_T; \mu) = -4C_F L_b. \quad (65)$$

At fixed ζ , Eq. (61) also gives the required one-loop renormalization-scale derivative,

$$\frac{\partial}{\partial \ln \mu} \tilde{C}_{q \leftarrow j}^{(1), \text{CSS2}} = [6C_F - 4C_F \ell_\zeta] \delta_{qj} \delta(1-z) - 4P_{qj}^{(0)}(z), \quad (66)$$

which is Eq. (76) of Collins and Rogers written with the standard leading-order DGLAP kernels of Eq. (54).

11.3 Momentum-space matching coefficients

The $\mathcal{K}_{\text{CSS2}}$ coefficient is the inverse transverse Fourier transform of Eq. (61). Using

$$L_b \mapsto \mathfrak{L}_0 = -\mathcal{L}_0, \quad L_b^2 \mapsto \mathfrak{L}_1 = -2\mathcal{L}_1, \quad (67)$$

we obtain the one-loop momentum-space kernels

$$\mathcal{C}_{q \leftarrow q}^{(1), \mathcal{K}_{\text{CSS2}}}(z, \mathbf{k}; \mu, \zeta) = C_F \left[-\frac{\pi^2}{6} \delta(1-z) + 2(1-z) \right] \delta^{(2)}(\mathbf{k}) \quad (68)$$

$$+ \left[2P_{qq}^{(0)}(z) - C_F(3 - 2\ell_\zeta) \delta(1-z) \right] \mathcal{L}_0(\mathbf{k}, \mu) \quad (69)$$

$$+ 2C_F \delta(1-z) \mathcal{L}_1(\mathbf{k}, \mu), \quad (70)$$

$$\mathcal{C}_{q \leftarrow g}^{(1), \mathcal{K}_{\text{CSS2}}}(z, \mathbf{k}; \mu, \zeta) = 2z(1-z) \delta^{(2)}(\mathbf{k}) + 2P_{qg}^{(0)}(z) \mathcal{L}_0(\mathbf{k}, \mu), \quad (71)$$

$$\mathcal{C}_{q \leftarrow q'}^{(1), \mathcal{K}_{\text{CSS2}}} = \mathcal{C}_{q \leftarrow \bar{q}}^{(1), \mathcal{K}_{\text{CSS2}}} = \mathcal{C}_{q \leftarrow \bar{q}'}^{(1), \mathcal{K}_{\text{CSS2}}} = 0. \quad (72)$$

The antiquark coefficients follow by charge conjugation. The gluon-initiated analogues for gluon TMDPDFs are not needed for the first fixed-target Drell-Yan validation, but the same map applies once the CSS2 gluon coefficients are inserted.

In generator form the same result is more compact:

$$\mathcal{C}_{q \leftarrow q}^{(1), \mathcal{K}_{\text{CSS2}}} = c_{q \leftarrow q}^{\text{can}}(z) \delta^{(2)} + \left[-2P_{qq}^{(0)}(z) + C_F(3 - 2\ell_\zeta) \delta(1-z) \right] \mathfrak{L}_0 - C_F \delta(1-z) \mathfrak{L}_1, \quad (73)$$

$$\mathcal{C}_{q \leftarrow g}^{(1), \mathcal{K}_{\text{CSS2}}} = c_{q \leftarrow g}^{\text{can}}(z) \delta^{(2)} - 2P_{qg}^{(0)}(z) \mathfrak{L}_0, \quad (74)$$

where

$$c_{q \leftarrow q}^{\text{can}}(z) = C_F \left[-\frac{\pi^2}{6} \delta(1-z) + 2(1-z) \right], \quad c_{q \leftarrow g}^{\text{can}}(z) = 2z(1-z). \quad (75)$$

Equations (70) and (71) are the first concrete large- k_T TMD-tail kernels for the project.

11.4 Momentum-space Collins-Soper kernel

With our sign convention in Eq. (14),

$$D_q^{(1),\text{CSS2}}(b_T; \mu) = -\widetilde{K}_q^{(1)}(b_T; \mu) = 4C_F L_b. \quad (76)$$

More generally,

$$D_i^{(1),\text{CSS2}}(b_T; \mu) = \Gamma_0^i L_b, \quad \Gamma_0^q = 4C_F, \quad \Gamma_0^g = 4C_A. \quad (77)$$

Therefore

$$\mathcal{D}_i^{(1),\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = \Gamma_0^i \mathcal{L}_0(\mathbf{k}; \mu) = -\Gamma_0^i \mathcal{L}_0(\mathbf{k}, \mu), \quad (78)$$

and

$$\mathcal{D}_q^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = -4C_F a_s(\mu) \mathcal{L}_0(\mathbf{k}, \mu) + \mathcal{O}(a_s^2). \quad (79)$$

Although Eq. (79) is negative away from $k_T = 0$ with our plus-basis convention, this is not a positivity problem: $\mathcal{D}_i$ is a renormalized rapidity evolution distribution, not a probability density.

The RG check is immediate. Since

$$\frac{d}{d \ln \mu} \mathcal{L}_0(\mathbf{k}, \mu) = -2\delta^{(2)}(\mathbf{k}), \quad (80)$$

Eq. (78) implies

$$\frac{d}{d \ln \mu} \mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = 2\Gamma_0^i a_s(\mu) \delta^{(2)}(\mathbf{k}) + \mathcal{O}(a_s^2) = \gamma_{K,0}^i a_s(\mu) \delta^{(2)}(\mathbf{k}) + \mathcal{O}(a_s^2), \quad (81)$$

with $\gamma_{K,0}^i = 2\Gamma_0^i$, as required by Eq. (34).

11.5 First fixed-order singular-spectrum check

At order a_s , the $\mathcal{K}_{\text{CSS2}}$ Drell-Yan W term follows by inserting Eqs. (70) and (71) into the two TMD factors in Eq. (16), adding the one-loop CSS2 hard factor, and keeping only terms through a_s . We use a reduced CSS2 hard factor with the Born charge e_q^2 pulled outside the flavor sum. At $\mu = Q$ it is [36]

$$\widehat{H}_{q\bar{q}}^{\text{DY},\text{CSS2}}(Q, Q) = 1 + a_s h_q^{(1)} + \mathcal{O}(a_s^2), \quad h_q^{(1)} = C_F \left(-16 + \frac{7\pi^2}{3} \right). \quad (82)$$

For the first validation we take

$$\mu = Q, \quad \zeta_A = \zeta_B = Q^2, \quad \ell_{\zeta_A} = \ell_{\zeta_B} = 0, \quad (83)$$

and evaluate all collinear PDFs at $\mu = Q$ unless explicitly stated otherwise.

It is useful to introduce the $\mathcal{L}_0$ and delta-function coefficient kernels

$$b_{q \leftarrow q}(z) \equiv 2P_{qq}^{(0)}(z) - 3C_F \delta(1-z), \quad b_{q \leftarrow g}(z) \equiv 2P_{qg}^{(0)}(z), \quad (84)$$

$$c_{q \leftarrow q}(z) \equiv C_F \left[-\frac{\pi^2}{6} \delta(1-z) + 2(1-z) \right], \quad c_{q \leftarrow g}(z) \equiv 2z(1-z), \quad (85)$$

with identical antiquark kernels by charge conjugation. Then define the leading luminosity

$$\Phi_q^{(0)} \equiv f_{q/A}(x_A, Q) f_{\bar{q}/B}(x_B, Q) + f_{\bar{q}/A}(x_A, Q) f_{q/B}(x_B, Q), \quad (86)$$

and the one-leg insertions

$$\Phi_q^{b,A} \equiv \left[b_{q \leftarrow q} \otimes_x f_{q/A} + b_{q \leftarrow g} \otimes_x f_{g/A} \right] (x_A) f_{\bar{q}/B}(x_B, Q) \quad (87)$$

$$+ \left[b_{q \leftarrow q} \otimes_x f_{\bar{q}/A} + b_{q \leftarrow g} \otimes_x f_{g/A} \right] (x_A) f_{q/B}(x_B, Q), \quad (88)$$

$$\Phi_q^{b,B} \equiv f_{q/A}(x_A, Q) \left[b_{q \leftarrow q} \otimes_x f_{\bar{q}/B} + b_{q \leftarrow g} \otimes_x f_{g/B} \right] (x_B) \quad (89)$$

$$+ f_{\bar{q}/A}(x_A, Q) \left[b_{q \leftarrow q} \otimes_x f_{q/B} + b_{q \leftarrow g} \otimes_x f_{g/B} \right] (x_B), \quad (90)$$

and similarly $\Phi_q^{c,A}$ and $\Phi_q^{c,B}$ by replacing b with c in Eqs. (88) and (90). In this notation the complete one-loop singular W term in the $d^2\mathbf{q}$ distribution is

$$W_{AB}^{(0+1),\text{sing}}(Q, y, \mathbf{q}) = \sum_q e_q^2 \left\{ \Phi_q^{(0)} \delta^{(2)}(\mathbf{q}) \right. \quad (91)$$

$$+ a_s(Q) \left[4C_F \Phi_q^{(0)} \mathcal{L}_1(\mathbf{q}, Q) + (\Phi_q^{b,A} + \Phi_q^{b,B}) \mathcal{L}_0(\mathbf{q}, Q) \right. \quad (92)$$

$$\left. + \left(h_q^{(1)} \Phi_q^{(0)} + \Phi_q^{c,A} + \Phi_q^{c,B} \right) \delta^{(2)}(\mathbf{q}) \right] \Big\} + \mathcal{O}(a_s^2). \quad (93)$$

Equation (93) is the first fully explicit object that the code should reproduce from pure k_T convolutions. The coefficient of $\mathcal{L}_1$ is fixed only by the two incoming quark TMDs,

$$2C_F + 2C_F = 4C_F, \quad (94)$$

while all flavor mixing at this order enters through the $\mathcal{L}_0$ and $\delta^{(2)}(\mathbf{q})$ one-leg insertions.

11.6 Relation to the usual CSS A1 and B1 check

A common source of confusion is the location of the one-loop B coefficient. In the fixed- $\mu = Q$ representation of Eq. (93), the endpoint piece $3C_F \delta(1-z)$ inside each $2P_{qq}^{(0)}$ cancels the explicit $-3C_F \delta(1-z)$ in $b_{q \leftarrow q}$. This does not mean that the usual CSS B coefficient has disappeared. It has been reshuffled by DGLAP evolution between a canonical $\mu_b = b_0/b_T$ representation and a fixed- Q PDF representation.

To see the standard logarithmic check most directly, keep the incoming PDFs at the canonical scale $\mu_b = b_0/b_T$ and expand the CSS2 evolved diagonal $q\bar{q}$ channel. Suppressing flavor convolutions and non-logarithmic qg channels, the b_T -space coefficient is

$$\widetilde{W}_{q\bar{q}}^{(1),\text{sing}}(b_T) = e_q^2 f_{q/A}(x_A, \mu_b) f_{\bar{q}/B}(x_B, \mu_b) \left[-2C_F L_b^2 + 6C_F L_b + h_q^{(1)} - \frac{\pi^2}{3} C_F \right] + (q \leftrightarrow \bar{q}), \quad (95)$$

where the $-\pi^2 C_F/3$ comes from the two one-loop CSS2 TMD matching coefficients. Using Eq. (67), this becomes

$$W_{q\bar{q}}^{(1),\text{sing}}(\mathbf{q}) = e_q^2 f_{q/A}(x_A, \mu_b) f_{\bar{q}/B}(x_B, \mu_b) \left[4C_F \mathcal{L}_1(\mathbf{q}, Q) - 6C_F \mathcal{L}_0(\mathbf{q}, Q) \right. \quad (96)$$

$$\left. + \left(h_q^{(1)} - \frac{\pi^2}{3} C_F \right) \delta^{(2)}(\mathbf{q}) \right] + (q \leftrightarrow \bar{q}). \quad (97)$$

For $q_T > 0$, converting from $d^2\mathbf{q}$ to dq_T^2 gives

$$\pi a_s [4C_F \mathcal{L}_1(\mathbf{q}, Q) - 6C_F \mathcal{L}_0(\mathbf{q}, Q)] = \frac{\alpha_s}{\pi} \frac{1}{q_T^2} \left[C_F \ln \frac{Q^2}{q_T^2} - \frac{3}{2} C_F \right], \quad (98)$$

which is the usual one-loop CSS check with $A_q^{(1)} = C_F$ and $B_q^{(1)} = -3C_F/2$ in the dq_T^2 spectrum [2, 5]. The implementation should verify both Eq. (93) and the canonical cross-check Eq. (98); the two are related by ordinary DGLAP evolution of the collinear PDFs.

11.7 Binned plus-distribution form

Because $\mathcal{L}_n$ and $\delta^{(2)}$ are distributions, the validation should be performed on bin integrals rather than pointwise values at the origin. Define an azimuthally integrated bin functional

$$\mathcal{B}_n(q_a, q_b; Q) \equiv \int_{q_a^2}^{q_b^2} dq_T^2 \pi \mathcal{L}_n(\mathbf{q}, Q), \quad \mathcal{B}_\delta(q_a, q_b) \equiv \int_{q_a^2}^{q_b^2} dq_T^2 \pi \delta^{(2)}(\mathbf{q}). \quad (99)$$

For bins that do not touch the origin, $0 < q_a < q_b$, the plus prescription is inactive and

$$\mathcal{B}_n(q_a, q_b; Q) = \frac{1}{n+1} \left[\ln^{n+1} \frac{Q^2}{q_a^2} - \ln^{n+1} \frac{Q^2}{q_b^2} \right], \quad \mathcal{B}_\delta(q_a, q_b) = 0. \quad (100)$$

For a cumulant bin that starts at the origin and has $q_b < Q$, the plus prescription gives

$$\mathcal{B}_n(0, q_b; Q) = -\frac{1}{n+1} \ln^{n+1} \frac{Q^2}{q_b^2}, \quad \mathcal{B}_\delta(0, q_b) = 1. \quad (101)$$

Therefore the binned one-loop singular prediction corresponding to Eq. (93) is

$$W_{AB, \text{bin}}^{(0+1), \text{sing}}(q_a, q_b) = \sum_q e_q^2 \left\{ \Phi_q^{(0)} \mathcal{B}_\delta(q_a, q_b) \right. \quad (102)$$

$$+ a_s(Q) \left[4C_F \Phi_q^{(0)} \mathcal{B}_1(q_a, q_b; Q) + (\Phi_q^{b,A} + \Phi_q^{b,B}) \mathcal{B}_0(q_a, q_b; Q) \right. \quad (103)$$

$$\left. + \left(h_q^{(1)} \Phi_q^{(0)} + \Phi_q^{c,A} + \Phi_q^{c,B} \right) \mathcal{B}_\delta(q_a, q_b) \right] \Big\} + \mathcal{O}(a_s^2). \quad (104)$$

This is the preferred unit-test expression. A pointwise comparison at $q_T > 0$ should agree with the ordinary-function part of Eq. (93), but it cannot validate the delta-function constants or the plus-prescription signs.

Working decision. For the first implementation, use Eqs. (70), (71), (79), and (104) as the reference one-loop $\mathcal{K}_{\text{CSS2}}$ test set. Any later child scheme must reproduce these results after applying its finite conversion factors.

11.8 Executable one-loop bin validator

Version 0.6 promotes Eq. (104) from a formal expression to an executable validation target. The validator is intentionally narrower than a physical cross-section code. It contains no PDFs, no Y term, no fiducial cuts, no resummation, and no nonperturbative model. Its purpose is to test the distribution algebra that all later implementations must inherit.

Let a bin partition be

$$0 = q_0 < q_1 < \cdots < q_N, \quad B_{n,r} \equiv \mathcal{B}_n(q_r, q_{r+1}; Q), \quad B_{\delta,r} \equiv \mathcal{B}_\delta(q_r, q_{r+1}). \quad (105)$$

The first validation layer consists of four algebraic tests.

Plus-null test. For each $n \geq 0$, the plus prescription requires

$$\mathcal{B}_n(0, Q; Q) = 0. \quad (106)$$

This is the most basic sign check for the zero bin. In particular, for $q_b < Q$,

$$\mathcal{B}_0(0, q_b; Q) = -\ln \frac{Q^2}{q_b^2}, \quad \mathcal{B}_1(0, q_b; Q) = -\frac{1}{2} \ln^2 \frac{Q^2}{q_b^2}, \quad (107)$$

so the singular plus-distribution contribution to the first bin is not positive by itself. This is expected: the zero-bin plus terms, the Born delta term, virtual constants, resummation, the nonsingular remainder, and experimental binning together define the physical prediction.

Additivity test. For any $0 \leq q_a < q_b < q_c$,

$$\mathcal{B}_n(q_a, q_c; Q) = \mathcal{B}_n(q_a, q_b; Q) + \mathcal{B}_n(q_b, q_c; Q), \quad \mathcal{B}_\delta(q_a, q_c) = \mathcal{B}_\delta(q_a, q_b) + \mathcal{B}_\delta(q_b, q_c). \quad (108)$$

This checks that the zero-bin prescription is compatible with ordinary binning and that a rebinning operation does not change the cumulant.

Off-origin ordinary-function test. For bins with $q_a > 0$, the plus prescription is inactive. The diagonal one-loop kernel must reduce to the ordinary integral

$$a_s [4C_F \mathcal{B}_1(q_a, q_b; Q) - 6C_F \mathcal{B}_0(q_a, q_b; Q)] = \frac{\alpha_s}{\pi} C_F \left[\mathcal{B}_1(q_a, q_b; Q) - \frac{3}{2} \mathcal{B}_0(q_a, q_b; Q) \right]. \quad (109)$$

This is the bin-integrated version of Eq. (98). It fixes the $a_s = \alpha_s/(4\pi)$ convention, the factor of π from $d^2\mathbf{q} = \pi dq_T^2$, and the normalization of $\mathcal{L}_n$.

Delta-support test. For a partition that contains the origin in exactly one bin,

$$\sum_{r=0}^{N-1} B_{\delta,r} = 1, \quad B_{\delta,r} = \begin{cases} 1, & q_r = 0, \\ 0, & q_r > 0. \end{cases} \quad (110)$$

This isolates the hard factor and the delta-supported pieces of the TMD matching coefficients from the ordinary $q_T > 0$ spectrum.

Combining these tests, the canonical diagonal one-loop test weight for a single $q\bar{q}$ luminosity is

$$S_r^{(1),\text{diag}} = a_s(Q) \left[4C_F B_{1,r} - 6C_F B_{0,r} + \left(h_q^{(1)} - \frac{\pi^2}{3} C_F \right) B_{\delta,r} \right]. \quad (111)$$

The term proportional to $h_q^{(1)} - \pi^2 C_F/3$ is present in the canonical $\mu = Q$ diagonal CSS2 check in Eq. (97). The full general flavor-vector test in Eq. (104) is implemented in Sec. 11.9 with analytic toy PDFs.

The accompanying script `kt_kcss_one_loop_validator.py` implements Eqs. (99)–(111). With the sample choice $Q = 8 \text{ GeV}$, $\alpha_s(Q) = 0.25$, and bin edges $(0, 0.25, 0.5, 1, 2, 4, 8) \text{ GeV}$, the tests give

$$\max_n |\mathcal{B}_n(0, Q; Q)| = 0, \quad \max |\Delta_{\text{additivity}}| = 0, \quad \max_{q_a > 0} |\Delta_{\text{offzero}}| \simeq 1.1 \times 10^{-16}. \quad (112)$$

These numbers are not phenomenological predictions; they are regression-test tolerances for the distribution implementation.

Working decision. The first executable validation target is now fixed: any pure- k_T implementation of one-loop $\mathcal{K}_{\text{CSS2}}$ must reproduce Eqs. (106), (108), (109), and (110) before adding PDFs, flavor mixing, resummation, or a Y term.

11.9 Flavor-vector toy-PDF x-convolution validator

Version 0.7 extends the diagonal bin test to the full flavor-vector expression in Eq. (104). The point is still not phenomenology. The new validator uses analytic toy PDFs only to test the longitudinal convolution algebra and the x -space plus prescription independently of the transverse plus prescription.

For a test PDF $f(x)$ define

$$\phi_f(z; x) \equiv \frac{1}{z} f\left(\frac{x}{z}\right), \quad \phi_f(1; x) = f(x), \quad (113)$$

and the ordinary regular-kernel convolution functional

$$\mathcal{R}_K[f](x) \equiv \int_x^1 \frac{dz}{z} K(z) f\left(\frac{x}{z}\right). \quad (114)$$

The basic endpoint-subtracted plus convolution is

$$\mathcal{P}[f](x) \equiv \int_x^1 dz \frac{\phi_f(z; x) - f(x)}{1 - z} + f(x) \ln(1 - x), \quad (115)$$

which implements

$$\mathcal{P}[f](x) = \left[\frac{1}{1 - z} \right]_+ \otimes_x f. \quad (116)$$

We use the compact convention in which the plus prescription acts on the full endpoint singular function,

$$P_{qq}^{(0)}(z) = C_F \left[\frac{1 + z^2}{1 - z} \right]_+, \quad (117)$$

which is equivalent to Eq. (54). In this convention the validator checks the decomposed identity

$$\left[P_{qq}^{(0)} \otimes_x f \right](x) = C_F \left\{ 2\mathcal{P}[f](x) - \mathcal{R}_{1+z}[f](x) + \frac{3}{2}f(x) \right\}. \quad (118)$$

Equivalently, a direct implementation of the full plus distribution gives

$$\begin{aligned} \left[P_{qq}^{(0)} \otimes_x f \right]_{\text{direct}}(x) &= C_F \left[\int_x^1 dz \frac{1 + z^2}{1 - z} (\phi_f(z; x) - f(x)) \right. \\ &\quad \left. + f(x) \left(2 \ln(1 - x) + x + \frac{x^2}{2} \right) \right]. \end{aligned} \quad (119)$$

Equations (118) and (119) must agree numerically. The same test also verifies the endpoint cancellation in the $\mathcal{L}_0$ coefficient kernel,

$$[b_{q \leftarrow q} \otimes_x f](x) = \left[(2P_{qq}^{(0)} - 3C_F \delta(1 - z)) \otimes_x f \right](x) = C_F \{ 4\mathcal{P}[f](x) - 2\mathcal{R}_{1+z}[f](x) \}. \quad (120)$$

The remaining one-loop kernels in the validation formula are regular or delta-supported:

$$[b_{q \leftarrow g} \otimes_x g](x) = 2T_F \mathcal{R}_{z^2 + (1-z)^2}[g](x), \quad (121)$$

$$[c_{q \leftarrow q} \otimes_x f](x) = C_F \left[-\frac{\pi^2}{6} f(x) + 2\mathcal{R}_{1-z}[f](x) \right], \quad (122)$$

$$[c_{q \leftarrow g} \otimes_x g](x) = \mathcal{R}_{2z(1-z)}[g](x). \quad (123)$$

The toy PDFs are beta-like analytic functions,

$$f_{i/h}^{\text{toy}}(x) = N_{i/h} x^{a_{i/h}} (1-x)^{b_{i/h}} (1+c_{i/h}x), \quad 0 < x < 1, \quad (124)$$

with independent parameter choices for the two incoming hadrons and for $u, \bar{u}, d, \bar{d}, s, \bar{s}, g$. The exact parameter values are stored in the validator JSON output; they are not intended to approximate a physical PDF set.

The second validation layer has four regression tests:

$$\Delta_P \equiv \max_{h,i,x} \left| \left[P_{qq}^{(0)} \otimes_x f_{i/h}^{\text{toy}} \right]_{\text{direct}}(x) - \left[P_{qq}^{(0)} \otimes_x f_{i/h}^{\text{toy}} \right]_{\text{decomp}}(x) \right|, \quad (125)$$

$$\Delta_b \equiv \max_{h,i,x} \left| \left[(2P_{qq}^{(0)} - 3C_F\delta) \otimes_x f_{i/h}^{\text{toy}} \right](x) - C_F \{ 4\mathcal{P}[f_{i/h}^{\text{toy}}] - 2\mathcal{R}_{1+z}[f_{i/h}^{\text{toy}}] \}(x) \right|, \quad (126)$$

$$\Delta_W \equiv \max_r \left| W_r^{\text{master}} - W_r^{\text{explicit leg}} \right|, \quad (127)$$

$$\Delta_{\text{add}} \equiv \max |W(q_a, q_c) - W(q_a, q_b) - W(q_b, q_c)|. \quad (128)$$

Here W_r^{master} uses the aggregated luminosities $\Phi_q^{(0)}, \Phi_q^{b,A}, \Phi_q^{b,B}, \Phi_q^{c,A}, \Phi_q^{c,B}$ in Eq. (104), while $W_r^{\text{explicit leg}}$ independently constructs the two partonic channels $q_A \bar{q}_B$ and $\bar{q}_A q_B$ from one-leg insertions before summing flavors and charges. This duplicate path is deliberate: it checks the flavor bookkeeping and the charge-conjugation assumptions in Eqs. (88)–(90).

The accompanying script `kt_kcss_toypdf_xconv_validator.py` implements Eqs. (115)–(128). With the sample choice $Q = 8 \text{ GeV}$, $\alpha_s(Q) = 0.25$, $x_A = 0.32$, $x_B = 0.21$, and bin edges $(0, 0.25, 0.5, 1, 2, 4, 8) \text{ GeV}$, it gives

$$\Delta_P \simeq 8.0 \times 10^{-14}, \quad \Delta_b \simeq 1.8 \times 10^{-15}, \quad (129)$$

$$\Delta_W \simeq 0, \quad \Delta_{\text{add}} \simeq 1.4 \times 10^{-17}. \quad (130)$$

These values are numerical integration and floating-point regression checks. They do not validate a physical cross section; they validate the algebraic separation of longitudinal plus distributions, transverse plus distributions, flavor-vector luminosities, and one-loop $\mathcal{K}_{\text{CSS2}}$ hard/matching constants.

Working decision. The second executable validation target is now fixed: before attaching physical PDFs, resummation, or a Y term, the code must pass the longitudinal plus-distribution tests in Eqs. (125)–(128) with analytic toy PDFs.

12 Generator convolution algebra

Version 0.8 promotes the formal generator identity in Eq. (41) into an explicit convolution table that can be used by the evolution code. This is the algebraic bridge from the fixed one-loop singular spectrum to a genuine convolution-exponential Sudakov kernel. The construction follows the distribution-space logic of Ebert and Tackmann [11]: one should solve the evolution problem in a basis of distributions, not by assigning pointwise values to singular functions at $k_T = 0$.

12.1 Exact algebra in the generator basis

The generator basis is defined by

$$\mathcal{F}_{k \rightarrow b}[\mathcal{G}_\eta] \equiv \int d^2\mathbf{k} e^{i\mathbf{b} \cdot \mathbf{k}} \mathcal{G}_\eta(\mathbf{k}; \mu) = e^{-\eta L_b}, \quad L_b = \ln \left(\frac{\mu^2 b_T^2 e^{2\gamma_E}}{4} \right). \quad (131)$$

Since transverse convolution becomes ordinary multiplication after the Fourier map,

$$\mathcal{G}_\eta \otimes_T \mathcal{G}_\xi = \mathcal{G}_{\eta+\xi}. \quad (132)$$

Using the derivative basis of Eq. (43),

$$\mathfrak{L}_n(\mathbf{k}; \mu) = (-1)^{n+1} \left. \frac{\partial^{n+1}}{\partial \eta^{n+1}} \mathcal{G}_\eta(\mathbf{k}; \mu) \right|_{\eta=0}, \quad \mathcal{F}_{k \rightarrow b}[\mathfrak{L}_n] = L_b^{n+1}, \quad (133)$$

the convolution algebra is particularly simple:

$$\boxed{\mathfrak{L}_m \otimes_T \mathfrak{L}_n = \mathfrak{L}_{m+n+1}.} \quad (134)$$

This follows either directly from the Fourier representation or by expanding both sides of Eq. (132) in powers of η and ξ . The ordinary delta function is the identity element,

$$\delta^{(2)}(\mathbf{k}) \otimes_T A = A \otimes_T \delta^{(2)}(\mathbf{k}) = A. \quad (135)$$

Working decision. The internal evolution engine should use Eq. (134) or the analytic generator $\mathcal{G}_\eta$ whenever possible. The conventional $\mathcal{L}_n$ table is mainly for comparison with fixed-order singular coefficients and for bin-level output.

12.2 Projection onto the conventional Ln basis

For fixed-order comparison, we still need the conventional basis

$$\mathcal{L}_n(\mathbf{k}, \mu) = \frac{1}{\pi \mu^2} \left[\frac{\mu^2}{k_T^2} \ln^n \left(\frac{\mu^2}{k_T^2} \right) \right]_+. \quad (136)$$

Let

$$P_n(X) \equiv \mathcal{F}_{k \rightarrow b}[\mathcal{L}_n], \quad X \equiv L_b. \quad (137)$$

The generator expansion gives a compact formula for the Fourier dictionary:

$$P_n(X) = (-1)^n n! [\eta^{n+1}] \frac{e^{-\eta X}}{h(\eta)}, \quad h(\eta) = e^{-2\gamma_E \eta} \frac{\Gamma(1-\eta)}{\Gamma(1+\eta)} = \exp \left[2 \sum_{r=1}^{\infty} \frac{\zeta_{2r+1}}{2r+1} \eta^{2r+1} \right]. \quad (138)$$

The first entries are

$$P_0(X) = -X, \quad (139)$$

$$P_1(X) = -\frac{1}{2} X^2, \quad (140)$$

$$P_2(X) = -\frac{1}{3} X^3 - \frac{4}{3} \zeta_3, \quad (141)$$

$$P_3(X) = -\frac{1}{4} X^4 - 4\zeta_3 X, \quad (142)$$

$$P_4(X) = -\frac{1}{5} X^5 - 8\zeta_3 X^2 - \frac{48}{5} \zeta_5. \quad (143)$$

Equivalently,

$$\mathfrak{L}_0 = -\mathcal{L}_0, \quad (144)$$

$$\mathfrak{L}_1 = -2\mathcal{L}_1, \quad (145)$$

$$\mathfrak{L}_2 = -3\mathcal{L}_2 - 4\zeta_3\delta^{(2)}, \quad (146)$$

$$\mathfrak{L}_3 = -4\mathcal{L}_3 + 16\zeta_3\mathcal{L}_0, \quad (147)$$

$$\mathfrak{L}_4 = -5\mathcal{L}_4 + 80\zeta_3\mathcal{L}_1 - 48\zeta_5\delta^{(2)}. \quad (148)$$

The appearance of $\zeta_3, \zeta_5, \dots$ is not new dynamics. These constants are a basis-conversion effect caused by packaging the Fourier-transform constants into $h(\eta)$.

12.3 Conventional-basis convolution table

The conventional table is obtained by multiplying $P_m(X)P_n(X)$ and projecting the result back onto $\{\delta^{(2)}, \mathcal{L}_0, \mathcal{L}_1, \dots\}$. The first entries are

$$\mathcal{L}_0 \otimes_T \mathcal{L}_0 = -2\mathcal{L}_1, \quad (149)$$

$$\mathcal{L}_0 \otimes_T \mathcal{L}_1 = -\frac{3}{2}\mathcal{L}_2 - 2\zeta_3\delta^{(2)}, \quad (150)$$

$$\mathcal{L}_0 \otimes_T \mathcal{L}_2 = -\frac{4}{3}\mathcal{L}_3 + 4\zeta_3\mathcal{L}_0, \quad (151)$$

$$\mathcal{L}_1 \otimes_T \mathcal{L}_1 = -\mathcal{L}_3 + 4\zeta_3\mathcal{L}_0, \quad (152)$$

$$\mathcal{L}_0 \otimes_T \mathcal{L}_3 = -\frac{5}{4}\mathcal{L}_4 + 12\zeta_3\mathcal{L}_1 - 12\zeta_5\delta^{(2)}, \quad (153)$$

$$\mathcal{L}_1 \otimes_T \mathcal{L}_2 = -\frac{5}{6}\mathcal{L}_4 + 12\zeta_3\mathcal{L}_1 - 8\zeta_5\delta^{(2)}. \quad (154)$$

The table is symmetric under $m \leftrightarrow n$. Higher entries should be generated by Eq. (138), not typed by hand.

12.4 First convolution-exponential check

The one-loop quark Collins-Soper kernel in Eq. (79) is proportional to $\mathcal{L}_0$. Therefore the first nontrivial rapidity-evolution Green function is already controlled by the generator algebra. For any scalar λ ,

$$\text{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \delta^{(2)} + \lambda\mathcal{L}_0 + \frac{\lambda^2}{2}\mathcal{L}_0 \otimes_T \mathcal{L}_0 + \frac{\lambda^3}{6}\mathcal{L}_0 \otimes_T \mathcal{L}_0 \otimes_T \mathcal{L}_0 + \dots \quad (155)$$

Since $\mathcal{F}[\mathcal{L}_0] = -L_b$, this exponential is exactly

$$\boxed{\text{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \mathcal{G}_\lambda.} \quad (156)$$

Expanding in the conventional basis gives

$$\begin{aligned} \text{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} &= \delta^{(2)} + \lambda\mathcal{L}_0 - \lambda^2\mathcal{L}_1 + \lambda^3\left(\frac{1}{2}\mathcal{L}_2 + \frac{2}{3}\zeta_3\delta^{(2)}\right) \\ &\quad + \lambda^4\left(-\frac{1}{6}\mathcal{L}_3 + \frac{2}{3}\zeta_3\mathcal{L}_0\right) + \mathcal{O}(\lambda^5). \end{aligned} \quad (157)$$

This check is useful because it verifies the signs in Eqs. (149)–(150). For example,

$$\mathcal{L}_0 \otimes_T \mathcal{L}_0 \otimes_T \mathcal{L}_0 = 3\mathcal{L}_2 + 4\zeta_3\delta^{(2)}, \quad (158)$$

which is the λ^3 coefficient of Eq. (157) multiplied by $3!$.

12.5 Executable generator-algebra validator

The accompanying script `kt_kcss_generator_convolution_validator.py` implements Eq. (138), generates the table through $\mathcal{L}_3 \otimes_T \mathcal{L}_3$, and checks three identities:

1. the exact generator-basis rule $\mathfrak{L}_m \otimes_T \mathfrak{L}_n = \mathfrak{L}_{m+n+1}$;
2. the conventional-basis projection by verifying $P_m(X)P_n(X) - \mathcal{F}[\mathcal{L}_m \otimes_T \mathcal{L}_n] = 0$;
3. the one-loop exponential identity $\text{Exp}_{\otimes_T}\{\lambda \mathcal{L}_0\} = \mathcal{G}_\lambda$ through the tested order.

The regression report gives

$$\text{passed} = \text{True}, \quad N_{\text{frak failures}} = 0, \quad \Delta_{\text{projection}} = 0, \quad \Delta_{\text{exp}} = 0. \quad (159)$$

This is the first validation step that directly targets the convolution-exponential structure needed for k_T -space Sudakov evolution.

Working decision. The generator algebra is now a validated component of the formalism. It supplies the engine used in Sec. 13 to build the first one-loop evolution Green function; the next extension is the running-coupling and two-loop version of $\mathcal{D}_i$ and $\mathcal{C}_{i \leftarrow j}$.

13 One-loop evolution Green function

Version 0.9 turns the generator algebra of Sec. 12 into the first explicit k_T -space evolution Green function. The purpose here is not yet to claim a phenomenological NLL implementation. The purpose is to validate the exact one-loop, fixed-coupling convolution dynamics that any higher-logarithmic implementation must reduce to. Running-coupling evolution, profile scales, and two-loop kernels are the next layer.

13.1 One-loop anomalous dimensions and variables

At one loop we write

$$\mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = a_s \Gamma_0^i \mathfrak{L}_0(\mathbf{k}; \mu) + \mathcal{O}(a_s^2) = -a_s \Gamma_0^i \mathcal{L}_0(\mathbf{k}, \mu) + \mathcal{O}(a_s^2), \quad (160)$$

and

$$\gamma_F^i(\mu, \zeta) = a_s \left[\gamma_{F,0}^i - \Gamma_0^i \ln \left(\frac{\zeta}{\mu^2} \right) \right] + \mathcal{O}(a_s^2). \quad (161)$$

For the quark TMD in the CSS2 convention used above,

$$\Gamma_0^q = 4C_F, \quad \gamma_{F,0}^q = 6C_F, \quad (162)$$

where the second equality follows from the coefficient-evolution check in Eq. (66). The gluon channel is obtained by replacing these constants by the corresponding CSS2 gluon anomalous-dimension coefficients.

For a path from (μ_i, ζ_i) to (μ_f, ζ_f) define

$$L_\mu \equiv \ln \frac{\mu_f}{\mu_i}, \quad \rho \equiv \ln \sqrt{\frac{\zeta_f}{\zeta_i}}, \quad \ell_i \equiv \ln \frac{\zeta_i}{\mu_i^2}, \quad \ell_f \equiv \ln \frac{\zeta_f}{\mu_f^2}. \quad (163)$$

These variables obey

$$\ell_f = \ell_i + 2\rho - 2L_\mu. \quad (164)$$

We hold a_s fixed in this subsection. This is an algebraic one-loop Green function; NLL evolution will replace the fixed a_s factors by RG integrals.

13.2 Rapidity evolution at fixed μ

At fixed μ , Eq. (32) gives

$$F_i(x, \mathbf{k}; \mu, \zeta_f) = \left[\mathcal{U}_i^\zeta(\mu; \zeta_f, \zeta_i) \otimes_T F_i(x; \mu, \zeta_i) \right] (\mathbf{k}), \quad (165)$$

with

$$\mathcal{U}_i^\zeta(\mathbf{k}; \mu; \zeta_f, \zeta_i) = \text{Exp}_{\otimes_T} \{ -\rho \mathcal{D}_i(\mu) \} = \text{Exp}_{\otimes_T} \left\{ a_s \Gamma_0^i \rho \mathcal{L}_0(\mu) \right\} + \mathcal{O}(a_s^2 \text{ in the kernel}). \quad (166)$$

Using Eq. (156), the fixed- μ rapidity Green function is therefore

$$\boxed{\mathcal{U}_i^\zeta(\mathbf{k}; \mu; \zeta_f, \zeta_i) = \mathcal{G}_{\eta_i}(\mathbf{k}; \mu), \quad \eta_i = a_s \Gamma_0^i \rho.} \quad (167)$$

In b_T space this is simply

$$\tilde{\mathcal{U}}_i^\zeta(b; \mu; \zeta_f, \zeta_i) = \exp \left[-a_s \Gamma_0^i \rho L_b(\mu) \right], \quad (168)$$

which solves

$$\frac{\partial}{\partial \rho} \tilde{\mathcal{U}}_i^\zeta = -a_s \Gamma_0^i \rho L_b(\mu) \tilde{\mathcal{U}}_i^\zeta. \quad (169)$$

13.3 Adding the μ leg

Choose the path that first evolves in μ at fixed ζ_i , then evolves in ζ at fixed μ_f . The μ -leg is local in transverse momentum, so it contributes a scalar factor. Integrating Eq. (161) gives

$$\begin{aligned} R_i^{(1)}(\mu_f, \mu_i; \zeta_i) &\equiv \int_{\ln \mu_i}^{\ln \mu_f} d \ln \mu a_s \left[\gamma_{F,0}^i - \Gamma_0^i \ln \left(\frac{\zeta_i}{\mu^2} \right) \right] \\ &= a_s L_\mu \left[\gamma_{F,0}^i - \Gamma_0^i (\ell_i - L_\mu) \right]. \end{aligned} \quad (170)$$

The resulting one-loop fixed-coupling Green function, represented with the final scale μ_f in the generator, is

$$\boxed{\mathcal{U}_i^{(1\ell)}(\mathbf{k}; \mu_f, \zeta_f; \mu_i, \zeta_i) = \exp \left\{ R_i^{(1)}(\mu_f, \mu_i; \zeta_i) \right\} \mathcal{G}_{\eta_i}(\mathbf{k}; \mu_f), \quad \eta_i = a_s \Gamma_0^i \rho.} \quad (171)$$

Equivalently, the b_T -space image is

$$\tilde{\mathcal{U}}_i^{(1\ell)} = \exp \left\{ a_s L_\mu \left[\gamma_{F,0}^i - \Gamma_0^i (\ell_i - L_\mu) \right] - a_s \Gamma_0^i \rho L_b(\mu_f) \right\}. \quad (172)$$

The corresponding fixed-order expansion in the conventional $\mathcal{L}_n$ basis is

$$\mathcal{U}_i^{(1\ell)} = \delta^{(2)} + a_s L_\mu \left[\gamma_{F,0}^i - \Gamma_0^i (\ell_i - L_\mu) \right] \delta^{(2)} + a_s \Gamma_0^i \rho \mathcal{L}_0(\mathbf{k}, \mu_f) + \mathcal{O}(a_s^2). \quad (173)$$

Equation (173) is the object that should appear when the resummed kernel is expanded to first order.

13.4 Path-independence check

The same endpoint evolution can be obtained by first evolving in ζ at μ_i and then evolving in μ at fixed ζ_f . That path gives

$$\mathcal{U}_{i,\zeta\text{-first}}^{(1\ell)} = \exp\left\{a_s L_\mu \left[\gamma_{F,0}^i - \Gamma_0^i(\ell_f + L_\mu)\right]\right\} \mathcal{G}_{\eta_i}(\mathbf{k}; \mu_i). \quad (174)$$

The two forms are identical. In b_T space the generator satisfies

$$L_b(\mu_f) = L_b(\mu_i) + 2L_\mu, \quad \mathcal{G}_\eta(\mu_i) = e^{2\eta L_\mu} \mathcal{G}_\eta(\mu_f), \quad (175)$$

while Eq. (164) relates ℓ_i and ℓ_f . These two relations give

$$\mathcal{U}_{i,\mu\text{-first}}^{(1\ell)} = \mathcal{U}_{i,\zeta\text{-first}}^{(1\ell)}. \quad (176)$$

This is the one-loop fixed-coupling version of path independence in the (μ, ζ) plane. It is the momentum-space analogue of the usual Collins-Soper/RG consistency condition.

13.5 Differential-equation and semigroup checks

The Green function in Eq. (171) satisfies the endpoint differential equations. Holding the initial point fixed,

$$\frac{\partial}{\partial \rho} \mathcal{U}_i^{(1\ell)} = -\mathcal{D}_i(\mu_f) \otimes_T \mathcal{U}_i^{(1\ell)} + \mathcal{O}(a_s^2 \text{ in the kernel}), \quad (177)$$

and

$$\frac{\partial}{\partial \ln \mu_f} \mathcal{U}_i^{(1\ell)} = \gamma_F^i(\mu_f, \zeta_f) \mathcal{U}_i^{(1\ell)} + \mathcal{O}(a_s^2 \text{ in the kernel}). \quad (178)$$

The second check uses both the explicit scalar derivative and the scale derivative of $\mathcal{G}_\eta(\mathbf{k}; \mu_f)$.

The Green function also obeys the semigroup property. For three points (μ_0, ζ_0) , (μ_1, ζ_1) , and (μ_2, ζ_2) ,

$$\mathcal{U}_i(2; 0) = \mathcal{U}_i(2; 1) \otimes_T \mathcal{U}_i(1; 0), \quad (179)$$

where the product is a transverse convolution in k_T space and ordinary multiplication in b_T space. This check will become nontrivial numerically once the kernel is used on a finite k_T grid.

13.6 Executable evolution-Green validator

The accompanying script `kt_kcss_one_loop_evolution_validator.py` validates Eqs. (171)–(179) symbolically in b_T space. It checks

1. equality of the μ -first and ζ -first paths;
2. the rapidity differential equation;
3. the μ differential equation at the final endpoint;
4. the semigroup law for three arbitrary points;
5. the $\mathcal{O}(a_s)$ conventional-basis expansion in Eq. (173).

The regression report gives

$$\text{passed} = \text{True}, \quad \Delta_{\text{path}} = 0, \quad \Delta_{\rho} = 0, \quad \Delta_{\mu} = 0, \quad \Delta_{\text{semi}} = 0. \quad (180)$$

For a quark-like smoke point with $\Gamma_0^q = 16/3$, $\gamma_{F,0}^q = 8$, and $a_s = 0.02$, the script finds

$$\eta_q \simeq 2.13 \times 10^{-2}, \quad R_q^{(1)} \simeq 3.52 \times 10^{-2}, \quad (181)$$

which is only a numerical smoke check, not a phenomenological prediction.

Working decision. The first evolution Green function is now fixed and validated at one-loop, fixed-coupling level. The next formal layer is to replace the fixed a_s factors by running-coupling RG integrals and then include the two-loop Collins-Soper kernel and matching coefficients needed for the first NLL/NNLL W -term prototype.

14 Two-loop Collins-Soper kernel and NNLO matching interface

Version 0.10 adds the first genuinely higher-order perturbative ingredient. The Collins-Soper kernel is compact enough to write explicitly in the $\mathcal{K}_{\text{CSS2}}$ basis. The full two-loop TMD matching coefficients are much longer; for them we define the import interface, scheme checks, and k_T projection map rather than duplicating pages of known coefficient functions. This is the right split for the code: the CS kernel is an internal analytic kernel, while NNLO matching tables should be imported from a vetted source and then transformed into our distribution basis.

14.1 Conventions and anomalous dimensions

We keep the expansion parameter

$$a_s(\mu) \equiv \frac{\alpha_s(\mu)}{4\pi}, \quad \frac{da_s}{d \ln \mu} = -2\beta_0 a_s^2 + \mathcal{O}(a_s^3), \quad \beta_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f. \quad (182)$$

For a parton in representation i , with $C_i = C_F$ for quarks and $C_i = C_A$ for gluons,

$$\Gamma_0^i = 4C_i, \quad (183)$$

$$\Gamma_1^i = 4C_i \left[\left(\frac{67}{9} - \frac{\pi^2}{3} \right) C_A - \frac{20}{9} T_F n_f \right]. \quad (184)$$

The two-loop boundary coefficient of the CSS2 Collins-Soper kernel, in the present normalization $D_i = -\widetilde{K}_i$, is

$$d_i^{(2)} = C_i \left[C_A \left(\frac{808}{27} - 28\zeta_3 \right) - \frac{224}{27} T_F n_f \right]. \quad (185)$$

For quarks this follows directly from the two-loop $\widetilde{K}$ result compiled by Collins and Rogers. Their Eq. (69) gives $\widetilde{K}_q$ through a_s^2 with $\ln(b_T \mu / (2e^{-\gamma_E})) = L_b/2$; after setting $D_q = -\widetilde{K}_q$ one obtains Eqs. (184)–(185). The same constant is equivalently the negative of the two-loop rapidity anomalous-dimension boundary term in the Li-Zhu normalization, after translating between $\widetilde{K}$ and D conventions [36, 15].

14.2 Two-loop kernel in bT space

Solving the RG equation

$$\frac{d}{d \ln \mu} D_i(b_T; \mu) = 2\Gamma_{\text{cusp}}^i(\alpha_s) = 2 \left[a_s \Gamma_0^i + a_s^2 \Gamma_1^i + \mathcal{O}(a_s^3) \right] \quad (186)$$

with $L_b = \ln(\mu^2 b_T^2 e^{2\gamma_E}/4)$ gives

$$D_i^{\text{CSS2}}(b_T; \mu) = a_s \Gamma_0^i L_b + a_s^2 \left[\frac{1}{2} \beta_0 \Gamma_0^i L_b^2 + \Gamma_1^i L_b + d_i^{(2)} \right] + \mathcal{O}(a_s^3). \quad (187)$$

The coefficient of L_b^2 is not independent. It is fixed by the running of a_s in the one-loop term. The coefficient of L_b is fixed by the two-loop cusp anomalous dimension. The only new constant at this order is $d_i^{(2)}$.

The RG check is useful enough to record explicitly. Using $dL_b/d \ln \mu = 2$ and Eq. (182),

$$\frac{d}{d \ln \mu} D_i^{\text{CSS2}}(b_T; \mu) = 2a_s \Gamma_0^i + a_s^2 \left[-2\beta_0 \Gamma_0^i L_b + 2\beta_0 \Gamma_0^i L_b + 2\Gamma_1^i \right] + \mathcal{O}(a_s^3) \quad (188)$$

$$= 2a_s \Gamma_0^i + 2a_s^2 \Gamma_1^i + \mathcal{O}(a_s^3). \quad (189)$$

14.3 Two-loop kernel in the generator and conventional kT bases

The $\mathcal{K}_{\text{CSS2}}$ momentum-space kernel is obtained by the same Fourier map used at one loop:

$$L_b^m \quad \longmapsto \quad \begin{cases} \delta^{(2)}(\mathbf{k}), & m = 0, \\ \mathfrak{L}_{m-1}(\mathbf{k}; \mu), & m \geq 1. \end{cases} \quad (190)$$

Therefore

$$\mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = a_s \Gamma_0^i \mathfrak{L}_0 + a_s^2 \left[\frac{1}{2} \beta_0 \Gamma_0^i \mathfrak{L}_1 + \Gamma_1^i \mathfrak{L}_0 + d_i^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_s^3). \quad (191)$$

Projecting onto the conventional plus basis with $\mathfrak{L}_0 = -\mathcal{L}_0$ and $\mathfrak{L}_1 = -2\mathcal{L}_1$ gives

$$\mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = -a_s \Gamma_0^i \mathcal{L}_0 + a_s^2 \left[-\beta_0 \Gamma_0^i \mathcal{L}_1 - \Gamma_1^i \mathcal{L}_0 + d_i^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_s^3). \quad (192)$$

This is the first nontrivial test of the generator basis. The $\mathfrak{L}$ form keeps the RG structure transparent, while the $\mathcal{L}_n$ form is the one to use for binned singular spectra.

In the $\mathfrak{L}$ basis, the scale derivative is

$$\frac{d}{d \ln \mu} \mathfrak{L}_n(\mathbf{k}; \mu) = 2(n+1) \mathfrak{L}_{n-1}(\mathbf{k}; \mu), \quad \mathfrak{L}_{-1} \equiv \delta^{(2)}, \quad (193)$$

so Eq. (191) obeys the momentum-space RG equation

$$\frac{d}{d \ln \mu} \mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = 2 \left[a_s \Gamma_0^i + a_s^2 \Gamma_1^i \right] \delta^{(2)}(\mathbf{k}) + \mathcal{O}(a_s^3). \quad (194)$$

14.4 NNLO matching coefficients: import architecture

The NNLO matching coefficients are needed for NNLL'-type accuracy and for a stable large- k_T perturbative tail. They are known in the literature; Gehrmann, Lübbert, and Yang give the two-loop transverse-PDF calculation and its scale-logarithm recursion, Echevarria, Scimemi, and Vladimirov provide NNLO unpolarized TMDPDF/TMDFF matching coefficients for all flavor combinations, and Luo, Yang, Zhu, and Zhu provide unpolarized quark and gluon TMDPDF and TMDFF matching coefficients through N^3 LO, including ancillary machine-readable files [13, 14, 19]. We should import those coefficients rather than hand-transcribing them.

For the $\mathcal{K}_{\text{CSS2}}$ engine, the required interface is the polynomial representation

$$\tilde{C}_{i \leftarrow j}^{(2), \text{CSS2}}(z, b_T; \mu, \zeta) = \sum_{m=0}^4 \tilde{c}_{i \leftarrow j}^{(2,m)}(z; \ell_\zeta) L_b^m, \quad \ell_\zeta = \ln \frac{\zeta}{\mu^2}. \quad (195)$$

The upper power $m = 4$ is allowed because the one-loop coefficient already contains L_b^2 terms. The momentum-space projection is then purely mechanical:

$$\boxed{C_{i \leftarrow j}^{(2), \mathcal{K}_{\text{CSS2}}}(z, \mathbf{k}; \mu, \zeta) = \tilde{c}_{i \leftarrow j}^{(2,0)}(z; \ell_\zeta) \delta^{(2)} + \sum_{m=1}^4 \tilde{c}_{i \leftarrow j}^{(2,m)}(z; \ell_\zeta) \mathfrak{L}_{m-1}(\mathbf{k}; \mu).} \quad (196)$$

For output in the conventional basis, use Eq. (148); explicitly,

$$\begin{aligned} C_{i \leftarrow j}^{(2), \mathcal{K}_{\text{CSS2}}} = & \left[\tilde{c}_{ij}^{(2,0)} - 4\zeta_3 \tilde{c}_{ij}^{(2,3)} \right] \delta^{(2)} - \left[\tilde{c}_{ij}^{(2,1)} - 16\zeta_3 \tilde{c}_{ij}^{(2,4)} \right] \mathcal{L}_0 - 2\tilde{c}_{ij}^{(2,2)} \mathcal{L}_1 \\ & - 3\tilde{c}_{ij}^{(2,3)} \mathcal{L}_2 - 4\tilde{c}_{ij}^{(2,4)} \mathcal{L}_3, \end{aligned} \quad (197)$$

where the common arguments $(z; \ell_\zeta)$ and $(\mathbf{k}; \mu)$ have been suppressed. This formula is an important implementation checkpoint: all zeta constants in Eq. (197) are basis-conversion constants, not new hard physics.

14.5 Logarithmic reconstruction of the NNLO matching coefficient

The finite functions $\tilde{c}_{i \leftarrow j}^{(2,0)}(z)$ must be imported from a vetted CSS2-compatible source. However, the L_b and ℓ_ζ dependence is not an independent input. It is fixed by the Collins-Soper equation, the μ -RG equation, the DGLAP kernels, and the two-loop kernel above. This gives a strong convention check on any imported NNLO table.

For a compact scalar-channel projection, suppress the flavor matrices and x -convolutions and write

$$\tilde{C} = 1 + a_s C_1 + a_s^2 C_2 + \mathcal{O}(a_s^3), \quad P_\mu = 4a_s P_0 + 8a_s^2 P_1 + \mathcal{O}(a_s^3), \quad (198)$$

with

$$D_1(L_b) = \Gamma_0 L_b, \quad (199)$$

$$D_2(L_b) = \frac{1}{2} \beta_0 \Gamma_0 L_b^2 + \Gamma_1 L_b + d^{(2)}, \quad (200)$$

$$\gamma_F(a_s, \ell_\zeta) = a_s(\gamma_0 - \Gamma_0 \ell_\zeta) + a_s^2(\gamma_1 - \Gamma_1 \ell_\zeta) + \mathcal{O}(a_s^3). \quad (201)$$

The one-loop coefficient may be written as

$$C_1(L_b, \ell_\zeta) = C_1^{[0]}(L_b) - \frac{1}{2} \Gamma_0 \ell_\zeta L_b, \quad C_1^{[0]}(L_b) = c_1 + \left(\frac{\gamma_0}{2} - 2P_0 \right) L_b - \frac{1}{4} \Gamma_0 L_b^2. \quad (202)$$

Then the NNLO logarithmic terms are reconstructed from the finite boundary $c_2 \equiv C_2(0, 0)$ by

$$C_2^{[0]}(L_b) = c_2 + \frac{1}{2} \int_0^{L_b} dL \left[\gamma_1 - D_2(L) - 8P_1 + (\gamma_0 + 2\beta_0)C_1^{[0]}(L) - D_1(L)C_1^{[0]}(L) - 4C_1^{[0]}(L)P_0 \right], \quad (203)$$

$$C_2(L_b, \ell_\zeta) = C_2^{[0]}(L_b) - \frac{\ell_\zeta}{2} \left[D_2(L_b) + D_1(L_b)C_1^{[0]}(L_b) \right] + \frac{\Gamma_0^2}{8} \ell_\zeta^2 L_b^2. \quad (204)$$

Equations (203) and (204) are not intended to replace the full NNLO coefficient tables. Their role is to reconstruct and validate all logarithmic terms once the finite boundary functions are imported.

The full flavor-vector form is obtained by restoring matrix multiplication under x -convolution. Thus P_0 and P_1 are DGLAP matrices, products such as $C_1^{[0]}P_0$ become $C_1^{[0]} \otimes_x P_0$, and the quantities $D_n^i, \Gamma_n^i, \gamma_n^i$ multiply the outgoing TMD row i . The ordering in x -space should not be changed when translating the scalar check into code.

The two evolution equations checked by this reconstruction are

$$2 \frac{\partial}{\partial \ell_\zeta} \tilde{C}_{i \leftarrow j} = -D_i \tilde{C}_{i \leftarrow j}, \quad (205)$$

$$\frac{d}{d \ln \mu} \tilde{C}_{i \leftarrow j} = \gamma_F^i \tilde{C}_{i \leftarrow j} - \sum_k \tilde{C}_{i \leftarrow k} \otimes_x (P_\mu)_{k \leftarrow j}. \quad (206)$$

Here $P_\mu = 4a_s P^{(0)} + 8a_s^2 P^{(1)} + \dots$ is the DGLAP kernel for evolution with respect to $\ln \mu$. The factors of four and eight follow from the standard $df/d \ln \mu^2 = (\alpha_s/2\pi)P^{(0)} \otimes f + (\alpha_s/2\pi)^2 P^{(1)} \otimes f + \dots$ convention together with $a_s = \alpha_s/(4\pi)$.

14.6 Flavor channels required at two loops

The import layer must be flavor-vector valued. For a quark TMD, the NNLO channels are schematically

$$q \leftarrow q, \quad q \leftarrow \bar{q}, \quad q \leftarrow q', \quad q \leftarrow \bar{q}', \quad q \leftarrow g, \quad (207)$$

with $q' \neq q$. For a gluon TMD, the corresponding channels are

$$g \leftarrow g, \quad g \leftarrow q, \quad g \leftarrow \bar{q}. \quad (208)$$

Drell-Yan fixed-target phenomenology will mostly stress the quark channels, but the formalism and code must carry the gluon entries from the start so that the same $\mathcal{K}_{\text{CSS2}}$ objects can be used for collider Drell-Yan, Higgs-like color-singlet production, SIDIS, and eventually e^+e^- fragmentation observables.

14.7 Validation gates before using NNLO matching in a fit

The two-loop matching interface should not be considered active until the following tests pass:

1. **Scheme test.** The imported coefficients must be in CSS2 or converted to CSS2 by explicit finite factors. If an SCET/RRG table is used, the conversion must be applied before Eq. (196).
2. **One-loop regression.** Applying the same import/projection machinery to the one-loop table must reproduce Eqs. (70) and (71).
3. **RG test.** The imported $\tilde{C}^{(2)}$ must satisfy the two-loop μ and ζ evolution equations with the kernel in Eq. (187) and the NLO DGLAP kernels.

4. **Distribution projection test.** The b_T -space polynomial and the k_T -space distribution representation must agree after transforming back to b_T space.
5. **Observable test.** The expanded NNLO singular W term must agree with the known small- q_T singular coefficients bin by bin before any Y term or nonperturbative model is attached.

Working decision. For the next implementation stage, the CS kernel is internal and analytic. The NNLO matching coefficients are external perturbative data with a strict import, conversion, and projection interface. This prevents transcription errors and keeps the $\mathcal{K}_{\text{CSS2}}$ formalism from becoming tied to a single paper’s coefficient notation.

14.8 Executable two-loop ingredient validator

The accompanying script `kt_kcss_two_loop_ingredients_validator.py` checks the compact CS-kernel part of this section and the scalar projection of the NNLO matching-interface equations. It verifies

1. equivalence between Collins-Rogers Eq. (69), rewritten with L_b , and Eq. (187);
2. the two-loop RG identity in Eq. (189);
3. the projection from Eq. (191) to Eq. (192);
4. Casimir scaling through two loops, $D_g/D_q = C_A/C_F$, for the perturbative coefficients;
5. the scalar-channel CS and μ evolution equations for the NNLO matching reconstruction, with the finite $c^{(2,0)}(z)$ table left as external input.

The regression report gives

$$\text{passed} = \text{True}, \quad \Delta_{K \rightarrow D} = 0, \quad \Delta_{\text{RG}} = 0, \quad \Delta_{\text{proj}} = 0, \quad \Delta_{\text{Casimir}} = 0, \quad \Delta_{C^{(2)} \text{ RG}} = 0. \quad (209)$$

15 Running-coupling evolution kernel

Version 0.11 promotes the fixed-coupling Green function of Sec. 13 to a running-coupling endpoint kernel. The goal is still formal and algorithmic: define the RG-integral object that the pure- k_T code will evaluate, expose the path-independence checks, and obtain the first two-loop rapidity-leg projection in the generator and conventional distribution bases.

15.1 General RG-integral form

We write

$$\Gamma_i(a_s) = \sum_{n \geq 0} a_s^{n+1} \Gamma_n^i, \quad \gamma_i(a_s) = \sum_{n \geq 0} a_s^{n+1} \gamma_n^i, \quad (210)$$

$$\frac{da_s}{d \ln \mu} = -2 \sum_{n \geq 0} \beta_n a_s^{n+2} = -2\beta_0 a_s^2 - 2\beta_1 a_s^3 + \dots. \quad (211)$$

Here Γ_i is the cusp anomalous dimension for the TMD parton representation and γ_i is the noncusp piece in

$$\gamma_F^i(\mu, \zeta) = \gamma_i(a_s(\mu)) - \Gamma_i(a_s(\mu)) \ln\left(\frac{\zeta}{\mu^2}\right). \quad (212)$$

For reference,

$$\beta_1 = \frac{34}{3}C_A^2 - \frac{20}{3}C_A T_F n_f - 4C_F T_F n_f. \quad (213)$$

The coefficient γ_1^i is a CSS2 scheme input. We keep it as an imported perturbative coefficient rather than hard-coding a convention before the complete CSS2 coefficient interface is active.

For endpoint evolution from (μ_i, ζ_i) to (μ_f, ζ_f) define

$$L \equiv \ln \frac{\mu_f}{\mu_i}, \quad \rho \equiv \ln \sqrt{\frac{\zeta_f}{\zeta_i}}, \quad \ell_i \equiv \ln \frac{\zeta_i}{\mu_i^2}. \quad (214)$$

It is useful to introduce the RG integrals

$$A_{\Gamma_i}(\mu_f, \mu_i) \equiv \int_{\ln \mu_i}^{\ln \mu_f} d \ln \mu \Gamma_i(a_s(\mu)), \quad (215)$$

$$A_{\gamma_i}(\mu_f, \mu_i) \equiv \int_{\ln \mu_i}^{\ln \mu_f} d \ln \mu \gamma_i(a_s(\mu)), \quad (216)$$

$$S_{\Gamma_i}(\mu_f, \mu_i) \equiv \int_{\ln \mu_i}^{\ln \mu_f} d \ln \mu \ln\left(\frac{\mu}{\mu_i}\right) \Gamma_i(a_s(\mu)). \quad (217)$$

Then the μ leg at fixed ζ_i gives the scalar exponent

$$\boxed{R_i(\mu_f, \mu_i; \zeta_i) = A_{\gamma_i} - \ell_i A_{\Gamma_i} + 2S_{\Gamma_i}}. \quad (218)$$

This reduces to Eq. (170) when a_s is fixed and only the one-loop anomalous dimensions are retained.

Using a μ -first path, the parent CSS2 image of the evolution kernel is

$$\boxed{\tilde{\mathcal{U}}_i(b_T; \mu_f, \zeta_f; \mu_i, \zeta_i) = \exp[R_i(\mu_f, \mu_i; \zeta_i) - \rho D_i(b_T; \mu_f)]}. \quad (219)$$

This equation is used as a proof and validation image. The implementation remains in k_T space by replacing the last factor with a transverse convolution exponential.

15.2 Path independence

The ζ -first path gives

$$\tilde{\mathcal{U}}_{i, \zeta\text{-first}} = \exp[-\rho D_i(b_T; \mu_i) + A_{\gamma_i} - (\ell_i + 2\rho)A_{\Gamma_i} + 2S_{\Gamma_i}]. \quad (220)$$

Equations (219) and (220) are equal through the same perturbative order because

$$D_i(b_T; \mu_f) - D_i(b_T; \mu_i) = 2A_{\Gamma_i}(\mu_f, \mu_i), \quad (221)$$

which is the integrated Collins-Soper-kernel RG equation $dD_i/d \ln \mu = 2\Gamma_i$. In k_T space this is a statement about convolution kernels. The kernels commute because their Fourier images are ordinary functions of L_b .

15.3 Momentum-space representation through two loops

Separate the $\mathcal{K}_{\text{CSS2}}$ kernel into its identity and plus-distribution pieces,

$$\mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) = D_i^\delta(a_s(\mu)) \delta^{(2)}(\mathbf{k}) + \sum_{n \geq 0} D_{i,n}(a_s(\mu)) \mathfrak{L}_n(\mathbf{k}; \mu). \quad (222)$$

Through two loops, Eq. (191) gives

$$D_i^\delta(a_f) = a_f^2 d_i^{(2)} + \mathcal{O}(a_f^3), \quad (223)$$

$$D_{i,0}(a_f) = a_f \Gamma_0^i + a_f^2 \Gamma_1^i + \mathcal{O}(a_f^3), \quad (224)$$

$$D_{i,1}(a_f) = \frac{1}{2} a_f^2 \beta_0 \Gamma_0^i + \mathcal{O}(a_f^3), \quad (225)$$

where $a_f \equiv a_s(\mu_f)$. It is useful to isolate the new two-loop generator associated with the L_b^2 term:

$$\mathcal{H}_\kappa(\mathbf{k}; \mu) \equiv \text{Exp}_{\otimes_T}[-\kappa \mathfrak{L}_1(\mathbf{k}; \mu)], \quad \mathcal{F}_{k \rightarrow b}[\mathcal{H}_\kappa] = \exp[-\kappa L_b^2]. \quad (226)$$

Then the rapidity leg through the two-loop CS kernel may be written as

$$\text{Exp}_{\otimes_T}[-\rho \mathcal{D}_i(\mu_f)] = e^{-\rho D_i^\delta(a_f)} \mathcal{H}_{\kappa_i}(\mu_f) \otimes_T \mathcal{G}_{\eta_i}(\mu_f) + \mathcal{O}(a_f^3 \text{ in } \mathcal{D}_i), \quad (227)$$

with

$$\eta_i = \rho D_{i,0}(a_f), \quad \kappa_i = \rho D_{i,1}(a_f). \quad (228)$$

No new scheme information is hidden in $\mathcal{H}_\kappa$; it is just the k_T -space representation of the L_b^2 term in the two-loop Collins-Soper kernel. The pure- k_T running-coupling kernel is therefore

$$\boxed{\mathcal{U}_i^{\text{run}}(\mathbf{k}; f, i) = \exp[R_i(\mu_f, \mu_i; \zeta_i) - \rho D_i^\delta(a_f)] \text{Exp}_{\otimes_T} \left\{ -\rho \sum_{n \geq 0} D_{i,n}(a_f) \mathfrak{L}_n(\mathbf{k}; \mu_f) \right\}. \quad (229)}$$

The expression is diagonal in the TMD parton representation. Flavor mixing enters in the matching coefficients and collinear PDFs, not in the Collins-Soper rapidity kernel.

Expanding the rapidity part of Eq. (229) through a_f^2 gives

$$\mathcal{U}_i^\zeta(\mathbf{k}; \mu_f) = \delta^{(2)} - a_f \rho \Gamma_0^i \mathfrak{L}_0 + a_f^2 \left[-\rho \Gamma_1^i \mathfrak{L}_0 + \left(\frac{1}{2} \rho^2 (\Gamma_0^i)^2 - \frac{1}{2} \rho \beta_0 \Gamma_0^i \right) \mathfrak{L}_1 - \rho d_i^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_f^3). \quad (230)$$

Projecting to the conventional basis gives

$$\mathcal{U}_i^\zeta(\mathbf{k}; \mu_f) = \delta^{(2)} + a_f \rho \Gamma_0^i \mathcal{L}_0 + a_f^2 \left[\rho \Gamma_1^i \mathcal{L}_0 + \left(\rho \beta_0 \Gamma_0^i - \rho^2 (\Gamma_0^i)^2 \right) \mathcal{L}_1 - \rho d_i^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_f^3). \quad (231)$$

This equation is a useful regression target for the binned distribution layer. The $\mathcal{L}_1$ term contains both the genuine two-loop L_b^2 contribution in D_i and the square of the one-loop rapidity kernel.

15.4 Expansion around the initial coupling

For algebraic checks it is convenient to expand around $a_i \equiv a_s(\mu_i)$. To $\mathcal{O}(a_i^2)$,

$$a_f = a_i - 2\beta_0 a_i^2 L + \mathcal{O}(a_i^3). \quad (232)$$

The RG integrals are

$$A_{\Gamma_i} = a_i \Gamma_0^i L + a_i^2 \left(\Gamma_1^i L - \beta_0 \Gamma_0^i L^2 \right) + \mathcal{O}(a_i^3), \quad (233)$$

$$A_{\gamma_i} = a_i \gamma_0^i L + a_i^2 \left(\gamma_1^i L - \beta_0 \gamma_0^i L^2 \right) + \mathcal{O}(a_i^3), \quad (234)$$

$$S_{\Gamma_i} = \frac{1}{2} a_i \Gamma_0^i L^2 + a_i^2 \left(\frac{1}{2} \Gamma_1^i L^2 - \frac{2}{3} \beta_0 \Gamma_0^i L^3 \right) + \mathcal{O}(a_i^3). \quad (235)$$

Thus

$$\begin{aligned} R_i &= a_i L \left(\gamma_0^i - \Gamma_0^i \ell_i \right) + a_i \Gamma_0^i L^2 \\ &\quad + a_i^2 \left[\gamma_1^i L - \beta_0 \gamma_0^i L^2 - \ell_i \left(\Gamma_1^i L - \beta_0 \Gamma_0^i L^2 \right) + \Gamma_1^i L^2 - \frac{4}{3} \beta_0 \Gamma_0^i L^3 \right] + \mathcal{O}(a_i^3). \end{aligned} \quad (236)$$

Setting $\beta_0 = \Gamma_1^i = \gamma_1^i = d_i^{(2)} = 0$ reduces the full endpoint exponent to

$$a_i L \left[\gamma_0^i - \Gamma_0^i (\ell_i - L) \right] - a_i \rho \Gamma_0^i L_b(\mu_f), \quad (237)$$

which is Eq. (172).

15.5 Executable running-coupling validator

The accompanying script `kt_kcss_running_coupling_evolution_validator.py` checks the $\mathcal{O}(a_i^2)$ algebra symbolically. It verifies

1. the fixed-coupling limit of the scalar exponent;
2. the integrated identity $D_i(\mu_f) - D_i(\mu_i) = 2A_{\Gamma_i}$;
3. equality of the μ -first and ζ -first paths;
4. the rapidity endpoint equation;
5. the μ endpoint equation;
6. the semigroup law for two consecutive endpoint evolutions;
7. the projection of Eq. (227) back to the two-loop b_T -space rapidity exponent.

The regression report gives

$$\begin{aligned} \text{passed} &= \text{True}, & \Delta_{\text{fixed}} &= 0, & \Delta_D &= 0, & \Delta_{\text{path}} &= 0, \\ \Delta_\rho &= 0, & \Delta_\mu &= 0, & \Delta_{\text{semi}} &= 0, & \Delta_{\text{HG}} &= 0. \end{aligned} \quad (238)$$

For the quark-like smoke point used by the script,

$$R_q \simeq 9.49 \times 10^{-2}, \quad \eta_q \simeq 2.14 \times 10^{-2}, \quad \kappa_q \simeq 1.12 \times 10^{-3}, \quad (239)$$

which is only a numerical regression check, not a phenomenological prediction. The coefficient β_1 is already part of the running-coupling interface, but it first contributes to this explicit initial-coupling expansion at $\mathcal{O}(a_i^3)$.

Working decision. The evolution layer is now formulated as a running-coupling k_T -space convolution kernel. Version 0.12 attaches this kernel to the first binned resummed W -term prototype and validates its one-loop fixed-order expansion.

16 Binned resummed W -term prototype

Version 0.12 is the first point where the perturbative ingredients are assembled into a forward-operator prototype. The purpose is deliberately narrow. We do not yet attach a nonperturbative core, a Y term, fiducial recoil corrections, or external NNLO matching tables. Instead, we define the scale map and the perturbative boundary blocks, then check that the fixed-order expansion of the profile-scale construction reproduces the canonical CSS2 singular spectrum bin by bin.

16.1 Endpoint scale map

For the first Drell–Yan prototype use the symmetric endpoint choice

$$\mu_H = Q, \quad \mu_f = Q, \quad \zeta_{A,f} = \zeta_{B,f} = Q^2, \quad \zeta_{A,f}\zeta_{B,f} = Q^4. \quad (240)$$

The perturbative TMD boundary is evaluated at a transverse profile scale

$$\mu_i = \mu_T, \quad \zeta_i = \mu_T^2, \quad \rho_T \equiv \ln \frac{Q}{\mu_T}, \quad \ell_i = \ln \frac{\zeta_i}{\mu_i^2} = 0. \quad (241)$$

For formal validation, μ_T is held fixed over the bins being tested. A phenomenology implementation may later use a smooth profile, for example

$$\mu_T(q; Q, \mu_0) = \min \left[Q, \sqrt{q^2 + \mu_0^2} \right], \quad (242)$$

but such bin-dependent choices must be handled with care because changing the profile from bin to bin can spoil exact distributional additivity. The safe validation gate is a constant auxiliary μ_T followed by explicit scale-variation tests.

16.2 Perturbative boundary block and evolved TMD leg

Define the perturbative boundary block

$$\mathbb{B}_{i/h}^{[m_C]}(x, \mathbf{k}; \mu_T) \equiv \sum_j \left[\mathcal{C}_{i \leftarrow j}^{[m_C], \mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu_T, \mu_T^2) \otimes_x f_{j/h}(\mu_T) \right](x), \quad (243)$$

where m_C is the matching order retained. The evolved perturbative TMD leg is

$$\mathbb{F}_{i/h}^{[N, m_C]}(x, \mathbf{k}; Q, \mu_T) = \mathcal{U}_i^{[N]}(\mathbf{k}; Q, Q^2; \mu_T, \mu_T^2) \otimes_T \mathbb{B}_{i/h}^{[m_C]}(x, \mathbf{k}; \mu_T). \quad (244)$$

Here N denotes the perturbative order of the evolution kernel and anomalous-dimension integrals. The first prototype uses the running-coupling kernel of Eq. (229) and the one-loop matching block of Sec. 11.

The binned W -term prototype is

$$W_{AB,r}^{\text{proto}}(Q, y) = \int_{q_a < |q| < q_b} d^2 \mathbf{q} H_{q\bar{q}}^{\text{CSS2}}(Q, Q) \sum_q e_q^2 \left[\mathbb{F}_{q/A} \otimes_T \mathbb{F}_{\bar{q}/B} + \mathbb{F}_{\bar{q}/A} \otimes_T \mathbb{F}_{q/B} \right](\mathbf{q}), \quad (245)$$

with $x_A = (Q/\sqrt{s})e^y$ and $x_B = (Q/\sqrt{s})e^{-y}$ understood. In the symmetric quark channel the two evolution kernels can be combined before the boundary convolution:

$$\begin{aligned} \mathbb{U}_q^{\text{pair}}(\mathbf{k}; Q, \mu_T) &\equiv \mathcal{U}_q(\mathbf{k}; Q, Q^2; \mu_T, \mu_T^2) \otimes_T \mathcal{U}_q(\mathbf{k}; Q, Q^2; \mu_T, \mu_T^2) \\ &= \exp \left[2R_q(Q, \mu_T; \mu_T^2) - 2\rho_T D_q^\delta(a_Q) \right] \text{Exp}_{\otimes_T} \left\{ -2\rho_T \sum_{n \geq 0} D_{q,n}(a_Q) \mathfrak{L}_n(\mathbf{k}; Q) \right\}. \end{aligned} \quad (246)$$

This is still a pure- k_T object: the expression is evaluated by generator algebra and then projected onto bin-integrated distributions.

16.3 One-loop fixed-order expansion gate

The first nontrivial check is independence of the auxiliary scale μ_T at $\mathcal{O}(a_s)$. Work in the parent Fourier image only as a proof device and define

$$X \equiv L_b(Q), \quad L \equiv \ln \frac{Q}{\mu_T}, \quad L_i \equiv L_b(\mu_T) = X - 2L. \quad (247)$$

Let P_A denote the action of the one-loop DGLAP kernel on the leg- A luminosity, and let c_A denote the finite $\delta^{(2)}$ matching contribution on that leg. The one-loop boundary coefficient in the diagonal quark channel is

$$C_A^{(1)}(\mu_T) = c_A + 3C_F L_i - C_F L_i^2 - 2P_A L_i. \quad (248)$$

Since our DGLAP convention is $d f / d \ln \mu = 4a_s P^{(0)} \otimes_x f + \dots$, expressing the boundary PDF at μ_T through a PDF at Q gives

$$f(\mu_T) = f(Q) - 4a_s L P^{(0)} \otimes_x f(Q) + \mathcal{O}(a_s^2). \quad (249)$$

The one-loop evolution factor for a quark leg from (μ_T, μ_T^2) to (Q, Q^2) is

$$E_q^{(1)} = 6C_F L + 4C_F L^2 - 4C_F L X. \quad (250)$$

Therefore

$$C_A^{(1)}(\mu_T) - 4L P_A + E_q^{(1)} = c_A + 3C_F X - C_F X^2 - 2P_A X, \quad (251)$$

so the auxiliary μ_T dependence cancels at the first fixed-order gate. For two Drell–Yan legs and one-loop hard coefficient $h_q^{(1)}$,

$$W_{q\bar{q},b}^{(1)}(X) = h_q^{(1)} + c_A + c_B + 6C_F X - 2C_F X^2 - 2(P_A + P_B)X. \quad (252)$$

Using $\mathcal{F}[\mathcal{L}_0] = -X$ and $\mathcal{F}[\mathcal{L}_1] = -X^2/2$, this maps back to

$$\boxed{W_{q\bar{q},k}^{(1)} = 4C_F \mathcal{L}_1 + [2(P_A + P_B) - 6C_F] \mathcal{L}_0 + \left[h_q^{(1)} + c_A + c_B \right] \delta^{(2)}}. \quad (253)$$

This is the same diagonal/splitting structure as the one-loop binned singular-spectrum master formula in Sec. 11, now derived from a profile-scale evolved boundary. Its bin action is

$$W_{q\bar{q},r}^{(1)} = 4C_F \mathcal{B}_{1,r} + [2(P_A + P_B) - 6C_F] \mathcal{B}_{0,r} + \left[h_q^{(1)} + c_A + c_B \right] \mathcal{B}_{\delta,r}. \quad (254)$$

16.4 Two-loop rapidity projection for the prototype

The NNLL-style bridge begins with the pair rapidity kernel. Let

$$\rho_{\text{pair}} = 2\rho_T. \quad (255)$$

Then the two-loop CS-kernel projection in the conventional basis is

$$\begin{aligned} \mathbb{U}_{q,\zeta}^{\text{pair}}(\mathbf{k}; Q, \mu_T) &= \delta^{(2)} + a_Q \rho_{\text{pair}} \Gamma_0^q \mathcal{L}_0 \\ &\quad + a_Q^2 \left[\rho_{\text{pair}} \Gamma_1^q \mathcal{L}_0 + \left(\rho_{\text{pair}} \beta_0 \Gamma_0^q - \rho_{\text{pair}}^2 (\Gamma_0^q)^2 \right) \mathcal{L}_1 - \rho_{\text{pair}} d_q^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_Q^3). \end{aligned} \quad (256)$$

This is not yet the complete $\mathcal{O}(a_s^2)$ Drell–Yan singular spectrum: completing that requires the two-loop hard factor, the imported NNLO matching coefficients, the $P^{(1)}$ PDF evolution term, and products of one-loop matching and evolution pieces. Equation (256) is the first validated kernel sub-block needed for that calculation.

16.5 Executable binned W-term prototype validator

The accompanying script `kt_kcss_binned_wterm_prototype_validator.py` checks the algebra above. It verifies

1. the one-leg profile-scale identity in Eq. (251);
2. the two-leg identity in Eq. (252);
3. the projection to the distribution coefficients in Eq. (253);
4. the two-loop pair rapidity projection in Eq. (256);
5. bin-level equality for several auxiliary μ_T values.

The regression report gives

$$\begin{aligned} \text{passed} &= \text{True}, & \Delta_{\text{leg}} &= 0, \quad \Delta_{\text{pair}} = 0, \quad \Delta_{\text{coeffs}} = 0, \quad \Delta_{\text{rapidity}} = 0, \\ \max_r \left| W_r^{\text{profile}} - W_r^{\text{canonical}} \right| &= 0 \quad \text{for the smoke-test bins.} \end{aligned} \quad (257)$$

The last check is not a physical prediction. It is a regression test that the binned distribution layer, the DGLAP normalization, and the profile-scale cancellation are all consistent before the NNLO matching import and Y -term are activated.

Working decision. Version 0.12 establishes the first binned pure- k_T resummed- W prototype. The next required gate is the complete $\mathcal{O}(a_s^2)$ singular expansion: import the CSS2-compatible NNLO matching tables, include $H^{(2)}$ and $P^{(1)}$, and check the result against the CSS2 b_T expansion before benchmarking any fixed-order Y term.

17 Perturbative matching onto collinear PDFs

For $k_T \gg \Lambda_{\text{QCD}}$, the TMD must match onto collinear PDFs [17, 19]:

$$F_{i/h}(x, \mathbf{k}; \mu, \zeta) = \sum_j \left[\mathcal{C}_{i \leftarrow j}(\mathbf{k}; \mu, \zeta) \otimes_x f_{j/h}(\mu) \right] (x) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{k_T^2}\right). \quad (258)$$

The matching coefficient has the distributional structure

$$\mathcal{C}_{i \leftarrow j}(z, \mathbf{k}; \mu, \zeta) = c_{i \leftarrow j}^\delta(z; \mu, \zeta) \delta^{(2)}(\mathbf{k}) + \sum_{n \geq 0} c_{i \leftarrow j}^n(z; \mu, \zeta) \mathcal{L}_n(\mathbf{k}, \mu) + \mathcal{R}_{i \leftarrow j}(z, \mathbf{k}; \mu, \zeta), \quad (259)$$

where $\mathcal{R}$ is less singular as $k_T \rightarrow 0$ but contributes to the perturbative tail.

A pure- k_T extraction must not ask a flexible nonperturbative model to reproduce Eq. (258). The large- k_T tail is a perturbative boundary condition.

17.1 TMD tail versus cross-section tail

There are two different tails, and they should not be merged conceptually. The *TMD* tail is the universal OPE in Eq. (258); it is controlled by $\mathcal{C}_{i \leftarrow j}^{[\mathcal{K}_{\text{CSS2}}]}$ and collinear PDFs. The *cross-section* large- q_T tail is the nonsingular fixed-order remainder,

$$Y_{AB}^{[\mathcal{K}_{\text{CSS2}}]} = \left[\frac{1}{\sigma_0} \frac{d\sigma}{dQ^2 dy d^2\mathbf{q}} \right]_{\text{FO}} - \left[W_{AB}^{[\mathcal{K}_{\text{CSS2}}], \text{resum}} \right]_{\text{FO exp.}}. \quad (260)$$

Codes such as DYTURBO and CuTe-MCFM/MCFM are appropriate for validating the fixed-order and matched cross-section side of Eq. (260) [37, 38, 39]. They do not replace the universal TMD matching coefficients in Eq. (258).

18 Model-agnostic nonperturbative module

The perturbative accuracy label belongs to the $\mathcal{K}_{\text{CSS2}}$ operator, not to a particular parameterization of long-distance physics. The nonperturbative input is therefore defined as a module,

$$\mathcal{M}_{\text{NP}} : (i, h, x, \mathbf{k}; \mu_0, \zeta_0, \theta) \mapsto \text{an admissible long-distance transverse structure}, \quad (261)$$

which may be supplied by a neural network, spline basis, Gaussian mixture, analytic model, lattice-informed input, renormalon-inspired ansatz, or any other controlled parameterization. A paper or code may choose one such realization, but the formal prescription does not require DNNs.

There are three useful module placements.

1. **Boundary-core module.** At a boundary scale (μ_0, ζ_0) ,

$$F_{i/h}^{\mathcal{K}_{\text{CSS2}}} = F_{i/h}^{\text{pert}} + F_{i/h}^{\text{core}}(\theta) - F_{i/h}^{\text{sub}}(\theta), \quad (262)$$

where $F^{\text{pert}} = \mathcal{C} \otimes_x f$ gives the perturbative tail and F^{sub} removes the overlap between the fitted core and the perturbative matching region.

2. **Smearing module.** The perturbative boundary is convolved with a normalized or regulated transverse smearing kernel,

$$F_{i/h}^{\mathcal{K}_{\text{CSS2}}} = S_{i/h}^{\text{NP}}(\theta) \otimes_T F_{i/h}^{\text{pert}} + F_{i/h}^{\text{power}}, \quad (263)$$

with the normalization of S^{NP} defined through regulated cumulants rather than an unregulated full- k_T probability condition.

3. **CS-kernel module.** Long-distance rapidity evolution is parametrized by an additive contribution to the Collins-Soper kernel,

$$\mathcal{D}_i = \mathcal{D}_i^{\text{pert}} + \mathcal{D}_i^{\text{NP}}(\theta), \quad (264)$$

with the perturbative part fixed by the $N^k\text{LL}$ ingredient manifest.

Any admissible module must satisfy the following minimal conditions.

1. It must not alter the perturbative ingredient content used to assign the $N^k\text{LL}'$ label.
2. It must be defined in the same flavor basis, scheme, and (μ, ζ) convention as the perturbative legs it modifies.
3. It must have well-defined actions on bins, cumulants, or smooth test functions.
4. It must not be asked to reproduce the perturbative large- k_T tail that is already fixed by $\mathcal{C}_{i \leftarrow j} \otimes_x f_j$.
5. It must include a stated overlap or matching prescription when it is combined additively with a perturbative tail.

Working decision. DNNs are a flexible implementation of $\mathcal{M}_{\text{NP}}$, not part of the formal definition of $\mathcal{K}_{\text{CSS2}}$ and not part of the perturbative accuracy label. The formal paper should present the module interface; a later extraction paper may choose a DNN realization and assign model, experimental, and replica uncertainties to it.

19 Regulated normalization and transverse moments

Once perturbative tails are present, an unregulated identity such as

$$\int d^2\mathbf{k} F_{i/h}(x, \mathbf{k}; Q) = f_{i/h}(x, Q) \quad (265)$$

is not a scheme-independent normalization condition. The renormalized k_T -space TMD is a distribution with a perturbative large- k_T tail, and full moments are UV sensitive. The correct objects are regulated integral functionals [22],

$$\mathcal{I}_w[F_i] \equiv \int d^2\mathbf{k} w(k_T; K) F_{i/h}(x, \mathbf{k}; \mu, \zeta), \quad (266)$$

where w may be a hard cumulant cutoff, a smooth profile, a bin window, or an experimental acceptance weight. For perturbative K the short-distance part has the matching form

$$\mathcal{I}_w[F_i] = \sum_j \left[Z_{i \leftarrow j}^w(K; \mu, \zeta) \otimes_x f_{j/h}(\mu) \right](x) + \mathcal{I}_w[F_i^{\text{NP}}] + \mathcal{O}(\Lambda_{\text{QCD}}^p / K^p). \quad (267)$$

This relation is the normalization constraint. The special Fernando-Keller-style condition of a normalized profile is recovered only after choosing a finite low-scale model space in which the perturbative tail and UV-sensitive cumulants have been removed or regulated.

A regulated transverse width is therefore defined by

$$\langle k_T^2 \rangle_w(x; \mu, \zeta) = \frac{\int d^2\mathbf{k} w_2(k_T; K) k_T^2 F_{i/h}(x, \mathbf{k}; \mu, \zeta)}{\int d^2\mathbf{k} w_0(k_T; K) F_{i/h}(x, \mathbf{k}; \mu, \zeta)}. \quad (268)$$

The weight functions and cutoff are part of the definition of the observable. This is what allows the same framework to be used in fixed-target kinematics, where Q and the available q_T range are modest, and in collider kinematics, where the perturbative tail, fiducial cuts, and Y term are essential.

For the formal paper the default recommendation is a hybrid numerical representation: singular perturbative kernels are integrated analytically over bins or smooth test functions, while ordinary nonperturbative modules are evaluated on a radial grid and convolved with analytic subtraction terms. This keeps distributional information exact and avoids treating plus distributions as pointwise functions.

20 Accuracy theorem and N3LL-prime ingredient map

Version 0.13 fixes the formal accuracy convention used by the note. The purpose is to make the eventual paper independent of any particular nonperturbative parametrization. A DNN, spline, Gaussian-mixture model, lattice-informed model, or analytic ansatz may supply long-distance information, but none of those choices defines the perturbative accuracy label. The label belongs to the perturbative forward operator.

Working decision. From this point onward the project target is written in standard notation as $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ for the singular TMD operator. A full-spectrum Drell–Yan prediction should only be labeled $N^3\text{LL}' + N^3\text{LO}$ after the corresponding Y term and fixed-order subtraction are implemented through $\mathcal{O}(\alpha_s^3)$.

20.1 Definition of NkLL-prime-KCSS

Let W^{CSS2} denote the parent CSS2 singular term evaluated with a specified set of hard functions, TMD matching coefficients, anomalous dimensions, DGLAP kernels, PDFs, and scale profiles. Let $W^{\mathcal{K}_{\text{CSS2}}}$ denote the direct momentum-space expression obtained by

$$L_b^m \mapsto \mathfrak{L}_{m-1}(\mathbf{k}; \mu) \quad (m \geq 1), \quad 1 \mapsto \delta^{(2)}(\mathbf{k}), \quad (269)$$

and by replacing ordinary products in b_T space by transverse convolutions in k_T space.

Definition. A calculation has $N^k\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ accuracy if:

1. the anomalous dimensions and running coupling are included to the same loop order as in the corresponding $N^k\text{LL}'$ CSS2 calculation;
2. the hard and TMD matching boundary functions are included through $\mathcal{O}(\alpha_s^k)$;
3. the k_T kernels are obtained from the CSS2 kernels by the analytic distribution map in Eq. (269);
4. every transverse convolution is evaluated as a distributional convolution, or by its action on bins/test functions;
5. if a matched full-spectrum result is quoted, the Y term is built from the same fixed-order expansion order as the claimed matched accuracy.

The unprimed $N^k\text{LL}_{\mathcal{K}_{\text{CSS2}}}$ convention uses the same evolution ingredients but keeps the hard and matching boundary functions one order lower, through $\mathcal{O}(\alpha_s^{k-1})$ for $k \geq 1$.

This is an ingredient-equivalence definition. It does not require a numerical Fourier-Bessel transform in the extraction code. The Fourier image is used only to prove and validate the operator identity.

20.2 Representation-equivalence theorem

Theorem. Assume that the parent CSS2 calculation and the $\mathcal{K}_{\text{CSS2}}$ calculation use the same renormalized perturbative ingredients, the same collinear PDFs, the same scale choices, and the same nonperturbative boundary module expressed in their respective representations. If the $\mathcal{K}_{\text{CSS2}}$ kernels are obtained by Eq. (269) and evolved by the generator convolution algebra, then for any infrared-safe bin or smooth test function φ ,

$$\int d^2\mathbf{q} \varphi(\mathbf{q}) W^{\mathcal{K}_{\text{CSS2}}, N^k \text{LL}'}(\mathbf{q}) = \int d^2\mathbf{q} \varphi(\mathbf{q}) W^{\text{CSS2}, N^k \text{LL}'}(\mathbf{q}), \quad (270)$$

up to the perturbative truncation and power corrections of the parent CSS2 calculation.

Proof sketch. The transform of the derivative basis satisfies

$$\mathcal{F}_{k \rightarrow b}[\mathfrak{L}_n(\mathbf{k}; \mu)] = L_b^{n+1}, \quad \mathcal{F}_{k \rightarrow b}[\delta^{(2)}(\mathbf{k})] = 1. \quad (271)$$

Therefore the convolution exponential obeys

$$\mathcal{F}_{k \rightarrow b} \left[\text{Exp}_{\otimes_T} \left\{ \sum_{n \geq 0} r_n \mathfrak{L}_n \right\} \right] = \exp \left\{ \sum_{n \geq 0} r_n L_b^{n+1} \right\}. \quad (272)$$

Products of two b_T -space TMDs map to transverse convolutions of the corresponding k_T -space TMDs. The hard function and collinear PDFs are unchanged by the representation map. Thus the two expressions have identical distributional Fourier images, and hence identical actions on bins and test functions. This is the same logic underlying the use of distribution-space evolution kernels for transverse-momentum resummation, but lifted from the color-singlet spectrum to the TMD-leg operator [11].

20.3 Generic counting table

The counting convention used in this note is summarized in Table 3. It follows the common SCET/CSS practice that the cusp anomalous dimension is needed one loop higher than the noncusp anomalous dimensions, while the prime means that fixed-order boundary functions are kept through the same order as the logarithmic label. Different codes sometimes vary in naming conventions, so a paper using this framework must state this table explicitly.

Table 3: Working logarithmic accuracy convention for the singular W operator. Here $k = 0, 1, 2, 3, \dots$ corresponds to LL, NLL, NNLL, N³LL, etc. The boundary order refers to hard functions and TMD matching coefficients.

Label	Γ_{cusp}	β function	noncusp γ_F	CS kernel / rapidity anomalous dimension	hard and TMD matching boundary
LL	1 loop	1 loop	none/tree	tree or RG-generated one-loop log	tree
NLL	2 loops	2 loops	1 loop	1 loop	tree
NLL'	2 loops	2 loops	1 loop	1 loop	1 loop

Label	Γ_{cusp}	β function	noncusp γ_F	CS kernel / rapidity anomalous dimension	hard and TMD matching boundary
NNLL	3 loops	3 loops	2 loops	2 loops	1 loop
NNLL'	3 loops	3 loops	2 loops	2 loops	2 loops
N ³ LL	4 loops	4 loops	3 loops	3 loops	2 loops
N ³ LL'	4 loops	4 loops	3 loops	3 loops	3 loops

The two-loop import interface in Eq. (195) generalizes directly: an n -loop matching coefficient contains powers up to L_b^{2n} . The $\mathcal{K}_{\text{CSS2}}$ import rule is therefore purely mechanical:

$$\tilde{C}_{i \leftarrow j}^{(n)}(z, L_b, \ell_\zeta) = \sum_{m=0}^{2n} \tilde{c}_{i \leftarrow j}^{(n,m)}(z; \ell_\zeta) L_b^m \implies \mathcal{C}_{i \leftarrow j}^{(n), \mathcal{K}_{\text{CSS2}}} = \tilde{c}^{(n,0)} \delta^{(2)} + \sum_{m=1}^{2n} \tilde{c}^{(n,m)} \mathfrak{L}_{m-1}. \quad (273)$$

No Bessel integral is required once the coefficient tables are expressed as polynomials in L_b .

20.4 Required ingredients for N3LL-prime-KCSS

Table 4 is the project manifest for the perturbative operator. The entries are phrased as requirements for the singular TMD operator; the final row adds the extra requirement for a matched full-spectrum Drell–Yan prediction.

Table 4: Ingredient manifest for a $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ singular TMD operator.

Ingredient	Required order	Role in the pure- k_T prescription
Cusp anomalous dimension Γ_{cusp}^i	4 loops, $\Gamma_0^i, \dots, \Gamma_3^i$	Controls the double-logarithmic part of the μ evolution and the RG evolution of the CS kernel. The four-loop cusp anomalous dimension is available in QCD [30].
QCD beta function	4 loops, $\beta_0, \dots, \beta_3$	Required for the running-coupling integrals A_Γ , A_γ , and S_Γ at N ³ LL accuracy.
TMD noncusp anomalous dimension γ_F^i	3 loops, $\gamma_{F,0}^i, \gamma_{F,1}^i, \gamma_{F,2}^i$	Enters the scalar endpoint factor multiplying the transverse convolution exponential.
Collins–Soper kernel / rapidity anomalous dimension D_i	3 loops, including $d_i^{(2)}$ and $d_i^{(3)}$ in the chosen CSS2 convention	Defines the transverse convolution generator in the rapidity leg. The three-loop rapidity anomalous dimension was extracted by Li and Zhu [15]; four-loop results provide the path beyond N ³ LL [20, 21].

Ingredient	Required order	Role in the pure- k_T prescription
Hard function $H_{ij \rightarrow F}$	3 loops for the relevant channel	Provides the short-distance virtual coefficient. For Drell–Yan and Higgs color-singlet channels, the needed three-loop hard functions are obtained from finite parts of the quark and gluon form factors [31].
TMD matching coefficients $\mathcal{C}_{i \leftarrow j}$ and, where needed, TMDFF coefficients	3 loops, i.e. N ³ LO boundary functions	Fix the perturbative large- k_T tail and the primed boundary. Unpolarized quark and gluon TMD-PDF/TMDFF matching coefficients through N ³ LO are available in machine-readable or ancillary form in the modern TMD literature [17, 19].
Collinear DGLAP kernels for x evolution inside the matching logs	$P^{(0)}, P^{(1)}, P^{(2)}$ for the N ³ LL' singular operator	Needed to reconstruct the μ dependence of the three-loop matching coefficients and to convert boundary PDFs between profile scales. The three-loop unpolarized splitting functions are the NNLO DGLAP kernels [32, 33].
Flavor matrix structure	Full quark, antiquark, and gluon matrix	The formalism is not a fixed-target DY ansatz. The matching and evolution act on parton flavor vectors, with process-specific hard functions and charge/color projectors.
Distribution basis	All $\mathfrak{L}_n$ required by the imported coefficients and the expanded convolution exponential	At three loops, matching polynomials can reach L_b^6 , hence the import map can require $\mathfrak{L}_5$ before convolutions with evolution are expanded.
Fixed-order subtraction for full-spectrum matching	$\mathcal{O}(\alpha_s^3)$ for an N ³ LL'+N ³ LO Drell–Yan spectrum	Not part of the universal TMD definition. It is required only when the prediction is extended beyond the singular region through $Y = d\sigma_{\text{FO}} - [W]_{\text{FO}}$. Modern Drell–Yan benchmarks already combine N ³ LL' resummation with available $\mathcal{O}(\alpha_s^3)$ fixed-order information [34].

The DGLAP entry should not be confused with the PDF-set order. For a self-contained N³LL' singular operator, $P^{(0)}$ through $P^{(2)}$ are sufficient to reconstruct the three-loop matching-scale logarithms. A phenomenological implementation may still use NNLO, approximate N³LO, or future full N³LO PDF evolution depending on the chosen PDF input and on whether one aims at an

approximate N⁴LL or full-spectrum N³LO benchmark.

20.5 What the prime does and does not mean

The prime in $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ means

$$\text{N}^3\text{LL evolution} + \text{three-loop hard and TMD matching boundary functions.} \quad (274)$$

It does *not* mean that the nonperturbative model is determined to third order, and it does not certify the quality of a DNN fit. The most precise statement is

$$W_{\text{theory}}^{\mathcal{K}_{\text{CSS2}}} = W_{\text{pert}}^{\mathcal{K}_{\text{CSS2}}, N^3\text{LL}'} \otimes_T \mathcal{M}_{\text{NP}} + \text{power corrections}, \quad (275)$$

where $\mathcal{M}_{\text{NP}}$ is a separately specified nonperturbative module. A DNN is one possible representation of $\mathcal{M}_{\text{NP}}$, but so are splines, Gaussian mixtures, renormalon-inspired models, lattice-informed inputs, or process-specific effective models.

20.6 Executable ingredient-manifest validator

The companion script `kt_kcss_accuracy_ingredient_validator.py` encodes the counting convention in Table 3. It checks that the generic rule gives the manifest orders in Table 4 for $N^3\text{LL}'$, namely

$$\Gamma_{\text{cusp}} : 4 \text{ loops}, \quad \beta : 4 \text{ loops}, \quad \gamma_F : 3 \text{ loops}, \quad D_i : 3 \text{ loops}, \quad H, C : 3 \text{ loops}, \quad P_{ij} : P^{(0)} \dots P^{(2)}. \quad (276)$$

This validator is intentionally simple. Its purpose is to prevent later sections and code from silently changing the project accuracy convention.

Working decision. The accuracy convention is now fixed. Version 0.14 adds the N³LO matching-import interface and manifest-driven kernel object needed to turn the convention into an executable perturbative operator.

21 N³LO matching-import interface and manifest-driven kernel object

Version 0.14 turns the accuracy manifest of Sec. 20 into an operator contract. The purpose is to make the high-order implementation reproducible without hand-transcribing multi-page coefficient functions into the text. The note should state the interface precisely, while the code should import vetted coefficient tables from external ancillary files or equivalent libraries. The relevant published inputs include the N³LO unpolarized TMDPDF/TMDFF matching calculations of Ebert–Mistlberger–Vita and Luo–Yang–Zhu–Zhu [17, 19, 18], plus the high-order rapidity anomalous-dimension inputs discussed in Refs. [15, 20, 21].

Working decision. The paper should not print the full N³LO matching tables. It should print the scheme, normalization, basis, and import map, and require that an implementation import machine-readable CSS2-compatible coefficient data satisfying the checks below.

21.1 Coefficient-table contract

An external matching table accepted by the $\mathcal{K}_{\text{CSS2}}$ importer must be convertible to the CSS2 parent convention before the k_T map is applied. The minimal metadata contract is

$$\mathbb{C}_{\text{in}} = \left\{ \begin{array}{l} \text{scheme, operator type, } a_s \text{ normalization,} \\ L_b \text{ convention, } \ell_\zeta \text{ convention,} \\ \text{flavor basis, coefficient terms} \end{array} \right\}. \quad (277)$$

The first implementation uses

$$a_s = \frac{\alpha_s}{4\pi}, \quad L_b = \ln\left(\frac{\mu^2 b_T^2}{b_0^2}\right), \quad \ell_\zeta = \ln\left(\frac{\zeta}{\mu^2}\right). \quad (278)$$

The table may be supplied in a full flavor-vector basis or in an equivalent singlet, non-singlet, and gluon basis, but the loader must expose a full parton-vector action on

$$\mathbf{f}_h = (g, u, \bar{u}, d, \bar{d}, s, \bar{s}, \dots)_h. \quad (279)$$

A source table in another TMD scheme is allowed only if a finite conversion kernel is provided:

$$\tilde{C}_{i \leftarrow j}^{\text{CSS2}} = \sum_a \tilde{Z}_{i \leftarrow a}^{\text{CSS2} \leftarrow S} \otimes_x \tilde{C}_{a \leftarrow j}^S. \quad (280)$$

Only after this conversion is the $\mathcal{K}_{\text{CSS2}}$ distribution map applied. This keeps the first paper's accuracy claim tied to the Collins/CSS2 parent scheme rather than to an implicit mixture of rapidity-regulator conventions.

For each loop order $n \leq 3$, the CSS2-compatible input must be expressible as

$$\tilde{C}_{i \leftarrow j}^{(n), \text{CSS2}}(z, L_b, \ell_\zeta) = \sum_{m=0}^{2n} \tilde{c}_{i \leftarrow j}^{(n,m)}(z; \ell_\zeta) L_b^m, \quad (281)$$

where the coefficients $\tilde{c}^{(n,m)}$ are still distributions or functions in z . They must not be flattened into ordinary numbers unless the action on an x -space test function has already been specified. The x -space plus prescriptions and the transverse plus prescriptions are independent layers.

21.2 Mechanical map into the KCSS2 basis

Define the import operator $\mathbf{l}_{\mathcal{K}_{\text{CSS2}}}$ by

$$\mathbf{l}_{\mathcal{K}_{\text{CSS2}}}[\tilde{c}(z; \ell_\zeta)] = \tilde{c}(z; \ell_\zeta) \delta^{(2)}(\mathbf{k}), \quad (282)$$

$$\mathbf{l}_{\mathcal{K}_{\text{CSS2}}}[\tilde{c}(z; \ell_\zeta) L_b^m] = \tilde{c}(z; \ell_\zeta) \mathfrak{L}_{m-1}(\mathbf{k}; \mu), \quad m \geq 1. \quad (283)$$

Then the imported three-loop matching operator is

$$\mathcal{C}_{i \leftarrow j}^{\mathcal{K}_{\text{CSS2}}, [3]} = \delta_{ij} \delta(1-z) \delta^{(2)} + \sum_{n=1}^3 a_s^n \left[\tilde{c}_{i \leftarrow j}^{(n,0)} \delta^{(2)} + \sum_{m=1}^{2n} \tilde{c}_{i \leftarrow j}^{(n,m)} \mathfrak{L}_{m-1} \right]. \quad (284)$$

At N³LO the largest logarithmic power is L_b^6 , so the matching import can require $\mathfrak{L}_5$. Subsequent convolutions with the evolution Green function can generate higher $\mathfrak{L}_n$ in fixed-order expansions, which is why the implementation should use the generator algebra rather than a hand-coded finite convolution table.

21.3 Manifest-driven N3LL-prime kernel object

The perturbative object assembled by the implementation is not a single function. It is a manifest-driven operator

$$\mathbb{K}_F^{N^3\text{LL}'} = \left[H_{ij \rightarrow F}^{[3]}, \mathcal{U}_i^{N^3\text{LL}}, \mathcal{C}_{i \leftarrow j}^{[3]}, P_{ij}^{(0,1,2)}, \mathcal{B}_{\text{dist}} \right]_{\mathcal{K}_{\text{CSS2}}}, \quad (285)$$

where $\mathcal{B}_{\text{dist}}$ denotes the distribution-basis engine for bins, cumulants, or test functions. The rapidity part of the N³LL kernel uses the three-loop CS kernel in the derivative basis,

$$\mathcal{D}_i^{[3]}(\mathbf{k}; \mu) = D_i^{\delta, [3]} \delta^{(2)} + D_{i,0}^{[3]} \mathfrak{L}_0 + D_{i,1}^{[3]} \mathfrak{L}_1 + D_{i,2}^{[3]} \mathfrak{L}_2, \quad (286)$$

with

$$D_i^{\delta, [3]} = a_s^2 d_i^{(2)} + a_s^3 d_i^{(3)}, \quad (287)$$

$$D_{i,0}^{[3]} = a_s \Gamma_0^i + a_s^2 \Gamma_1^i + a_s^3 \left(\Gamma_2^i + 2\beta_0 d_i^{(2)} \right), \quad (288)$$

$$D_{i,1}^{[3]} = a_s^2 \frac{\beta_0 \Gamma_0^i}{2} + a_s^3 \left(\beta_0 \Gamma_1^i + \frac{\beta_1 \Gamma_0^i}{2} \right), \quad (289)$$

$$D_{i,2}^{[3]} = a_s^3 \frac{\beta_0^2 \Gamma_0^i}{3}. \quad (290)$$

These coefficients follow from the RG equation

$$\frac{d}{d \ln \mu} D_i(b_T; \mu) = 2\Gamma_i(\alpha_s) \quad (291)$$

with $da_s/d \ln \mu = -2\beta_0 a_s^2 - 2\beta_1 a_s^3 + \mathcal{O}(a_s^4)$. In particular, the $2\beta_0 d_i^{(2)}$ term in Eq. (288) is required by running-coupling consistency; omitting it would break path independence at $\mathcal{O}(a_s^3)$.

The endpoint evolution kernel is then represented as

$$\mathcal{U}_i^{N^3\text{LL}}(\mathbf{k}; f, i) = \exp \left[R_i^{[3]}(f, i) - \rho D_i^{\delta, [3]}(a_f) \right] \text{Exp}_{\otimes_T} \left\{ -\rho \sum_{r=0}^2 D_{i,r}^{[3]}(a_f) \mathfrak{L}_r(\mathbf{k}; \mu_f) \right\}, \quad (292)$$

where $R_i^{[3]}$ is built from $\Gamma_{0\dots 3}$, $\gamma_{F,0\dots 2}$, and $\beta_{0\dots 3}$ through the RG integrals in Sec. 15. Equation (292) is the direct k_T -space replacement for the local CSS2 Sudakov factor.

21.4 Assembly of the perturbative W operator

At a perturbative boundary scale (μ_T, ζ_T) , the imported matching coefficients construct the boundary block

$$\mathbb{B}_{i/h}^{[3]}(x, \mathbf{k}; \mu_T, \zeta_T) = \sum_j \left[\mathcal{C}_{i \leftarrow j}^{\mathcal{K}_{\text{CSS2}}, [3]}(\mathbf{k}; \mu_T, \zeta_T) \otimes_x f_{j/h}(\mu_T) \right](x). \quad (293)$$

The perturbative TMD leg is

$$\mathbb{F}_{i/h}^{\text{pert}, [3]}(x, \mathbf{k}; Q) = \mathcal{U}_i^{N^3\text{LL}}(\mathbf{k}; Q, Q^2; \mu_T, \zeta_T) \otimes_T \mathbb{B}_{i/h}^{[3]}(x, \mathbf{k}; \mu_T, \zeta_T). \quad (294)$$

For a generic color-singlet process $AB \rightarrow F$,

$$W_{AB \rightarrow F}^{\mathcal{K}_{\text{CSS2}}, N^3\text{LL}'}(\mathbf{q}) = \sum_{ij} H_{ij \rightarrow F}^{[3]}(Q, \mu_H) \left[\mathbb{F}_{i/A}^{\text{pert}, [3]} \otimes_T \mathbb{F}_{j/B}^{\text{pert}, [3]} \right](\mathbf{q}), \quad (295)$$

with the obvious replacement of one or both TMDPDF legs by TMDFF legs for SIDIS or back-to-back hadron production in e^+e^- . A nonperturbative module may be convolved with the perturbative leg or supplied as a boundary correction, but it is not part of the N^3LL' accuracy label:

$$\mathbb{F}_{i/h} = \mathbb{F}_{i/h}^{\text{pert},[3]} \otimes_T \mathcal{M}_{i/h}^{\text{NP}} + \text{power-suppressed or overlap-subtracted terms.} \quad (296)$$

This is the desired separation for the eventual paper: the perturbative prescription is generic, while DNNs, splines, analytic ansatzes, and lattice-informed inputs are choices for $\mathcal{M}_{\text{NP}}$.

21.5 Import and kernel validation gates

Before any extracted TMD is assigned an N^3LL' label, the following gates must pass:

1. **Metadata gate:** the coefficient source must specify its scheme, coupling normalization, L_b , ℓ_ζ , operator type, and flavor basis.
2. **Scheme gate:** if the source is not CSS2, the finite conversion in Eq. (280) must be supplied and tested.
3. **Power gate:** an n -loop coefficient may contain no power higher than L_b^{2n} , and L_b^6 must map to $\mathfrak{L}_5$ at $N^3\text{LO}$.
4. **Flavor gate:** the loaded basis must act on the full quark, antiquark, and gluon vector, even if a first Drell–Yan application uses only a subset of channels.
5. **RG gate:** the logarithmic coefficients must satisfy the CS, μ -RG, and DGLAP reconstruction equations through $\mathcal{O}(a_s^3)$.
6. **Distribution gate:** transverse convolutions must be validated by generator algebra and by bin/test-function actions, not by pointwise evaluation at $q_T = 0$.
7. **Expansion gate:** the fixed-order expansion of Eq. (295) through $\mathcal{O}(a_s^3)$ must agree with the equivalent CSS2 singular expansion bin by bin.

The companion script `kt_kcss_n3lo_matching_import_validator.py` implements a symbolic version of these checks. It builds a mock CSS2-compatible three-loop coefficient table, maps L_b^m to $\delta^{(2)}$ or $\mathfrak{L}_{m-1}$, constructs a manifest-driven N^3LL' kernel object, and verifies the three-loop CS-kernel RG identity. The mock coefficients are placeholders; the point of the validator is the interface and algebra, not phenomenology.

Working decision. After v0.15, the next nontrivial physics task is to acquire the raw ancillary files into the local coefficient directory, convert the Mathematica tables into the canonical coefficient schema, and run the $\mathcal{O}(a_s^2)$ and $\mathcal{O}(a_s^3)$ singular-spectrum gates with physical coefficients.

22 Physical high-order coefficient import gate

Section 21 defined the abstract coefficient-table contract and used a synthetic payload only to validate the map

$$L_b^0 \mapsto \delta^{(2)}(\mathbf{k}), \quad L_b^m \mapsto \mathfrak{L}_{m-1}(\mathbf{k}; \mu), \quad m \geq 1. \quad (297)$$

That test is necessary, but it is not physics input. Version 0.15 introduced the hard gate that separates a mock interface test from a source-stamped perturbative coefficient package. A run may

not advertise an $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ boundary unless the imported three-loop matching payload carries provenance, convention metadata, a scheme map to CSS2, and a successful fixed-order expansion check.

Working decision. Synthetic coefficient tables are allowed only for importer regression tests. They are not allowed to satisfy the hard, matching, or expansion requirements of an $N^3\text{LL}'$ physics claim. If the raw ancillary files or a verified normalized export are absent, the physical import gate must remain blocked.

22.1 Registered source families

The first source registry is deliberately conservative. It contains the NNLO analytic reference used for lower-order checks and the two public $N^3\text{LO}$ source families needed for the unpolarized quark, antiquark, and gluon matching interface.

Source key	Order and legs	Local payload status	Role in the $\mathcal{K}_{\text{CSS2}}$ importer
ESV2016	NNLO PDF/FF	paper-level reference	Analytic NNLO reference for unpolarized TMDPDF and TMDFF matching in all flavor combinations; used for the $\mathcal{O}(a_s^2)$ gate and lower-order checks of any $N^3\text{LO}$ import.
EMV2020	$N^3\text{LO}$ PDF	ancillary files expected	Independent $N^3\text{LO}$ TMDPDF/beam-function source for quark and gluon initial-state channels. Registered files include TMDPDF.m, PTBF.m, PTBF_ZExpansion.m, PTBF_ZbExpansion.m, and soft-function files.
LYZZ2021	$N^3\text{LO}$ PDF/FF	ancillary files expected	Primary all-leg unpolarized matching source for PDFs and FFs. Registered files include TMDPDF.m, TMDPDFN.m, BeamFunction.m, BeamfunctionN.m, TMDFF.m, TMDFFN.m, FFMatchingKernels.m, FFMatchingKernelsN.m, ancillary_readme.pdf, and softfunction.m.

This registry is not itself a coefficient table. It is a provenance and convention layer that decides which external payloads are eligible to be converted into canonical coefficient records. The public arXiv records for EMV2020 and LYZZ2021 list the relevant ancillary files, while the publications describe the $N^3\text{LO}$ unpolarized TMDPDF/TMDFF matching calculations [17, 19].

22.2 Source stamp and hash protocol

Each physical coefficient payload must carry a source stamp

$$S_C = \{\text{source, arXiv version, journal record, file name, SHA256, parser version, scheme map}\}. \quad (298)$$

The importer must reject a payload if

$$\text{source} \in \{\text{mock, synthetic, placeholder, toy}\} \quad (299)$$

or if the source is not convertible to the CSS2 parent convention. In other words, the v0.14 mock table validates the map in Eq. (297), but only a source-stamped physical table validates the perturbative content.

The accepted coefficient package should use a local read-only layout such as

```
coefficients/
  2012.03256/
    TMDPDF.m
    TMDPDFN.m
    BeamFunction.m
    BeamfunctionN.m
    ancillary_readme.pdf
    ...
  2006.05329/
    TMDPDF.m
    PTBF.m
    PTBF_ZExpansion.m
    PTBF_ZbExpansion.m
    info.txt
    ...
```

A released analysis should archive the exact source package or a normalized export with hashes. Live network access should not be required to reproduce a fit.

22.3 Canonical record schema

Most public high-order matching tables are distributed as Mathematica ancillary files. The extraction code should not evaluate arbitrary Mathematica programs during a fit. The preferred contract is a two-stage conversion,

$$*.m \text{ source} \longrightarrow \text{canonical JSON/YAML coefficient table} \longrightarrow \mathcal{K}_{\text{CSS2}} \text{ operator}. \quad (300)$$

The canonical table exposes records

$$\mathcal{R}_C = \{n, \text{object}, i \leftarrow j, m, r, c_{i \leftarrow j}^{(n, m, r)}(z), S_C\}, \quad (301)$$

where n is the loop order, m is the power of L_b^m , r labels rapidity-logarithm structure such as powers of ℓ_ζ , and S_C is the source stamp. The z -space object may remain symbolic, provided the importer can apply it to test functions and can check DGLAP and CS logarithmic recursions.

22.4 Acceptance predicate

The accepted physical payload for an $N^3\text{LL}'$ claim must satisfy

$$\begin{aligned} \text{Accept}_{N^3\text{LL}'} &= \text{Source} \wedge \text{Hash} \wedge \text{NotMock} \wedge \text{Scheme} \wedge \text{Power} \wedge \text{Flavor} \\ &\quad \wedge \text{RG} \wedge \text{Expansion}. \end{aligned} \quad (302)$$

The individual gates are:

1. **Source/hash:** the raw or normalized file has a documented source and SHA256 hash;
2. **NotMock:** the source is not synthetic or placeholder data;
3. **Scheme:** the payload is already CSS2 or has an explicit finite conversion to CSS2;
4. **Power:** an n -loop coefficient has no power higher than L_b^{2n} ;
5. **Flavor:** the flavor basis covers the process class being claimed;
6. **RG:** the logarithmic terms satisfy the CS, μ -RG, and DGLAP reconstruction equations;
7. **Expansion:** the fixed-order expansion of the imported W operator agrees with the reference singular spectrum through the claimed order.

22.5 NNLO then N3LO staged replacement

The physical replacement of the mock payload should be staged:

1. **NNLO gate:** import the vetted two-loop coefficient package and verify the $\mathcal{O}(a_s^2)$ singular W expansion, including all one-loop products, $H^{(1)}C^{(1)}$, $C^{(1)} \otimes_T C^{(1)}$, $P^{(1)}$, $D^{(2)}$, and $H^{(2)}$ terms.
2. **N³LO gate:** import the vetted three-loop coefficient package and verify the $\mathcal{O}(a_s^3)$ singular expansion, including $H^{(3)}$, $C^{(3)}$, three-loop DGLAP, three-loop CS evolution, and all lower-order products generated by the convolution exponential.

The expansion check is a binned or test-function equality,

$$\int_{\text{bin}} d^2 \mathbf{q} \left[W_{\mathcal{K}_{\text{CSS2}}}^{\text{imported}} - W_{\text{CSS2}}^{\text{singular}} \right] \alpha_{(a_s^n)} = 0, \quad n = 2, 3, \quad (303)$$

for every channel, flavor basis element, and chosen test luminosity.

22.6 Executable v0.15 audit status

The companion script `kt_kcss_physical_coeff_import_validator.py` implements the source-level audit. It does not load the v0.14 synthetic coefficient payload. It checks whether the expected EMV2020 and LYZZ2021 ancillary files are present in the local coefficient staging directory and records whether the higher-order expansion gate can be enabled.

In the v0.15 sandbox run the raw ancillary files were not locally available, so the audit status is

Source registry: defined. Mock payload: disabled for physics. Raw coefficient files: not locally present. Physical NNLO/N³LO expansion gates: specified but not yet executed.

This is a deliberate hard stop. It prevents the note from overclaiming that physical three-loop matching coefficients have been imported when only the interface algebra has been tested.

23 N3LL-prime label certificate

The previous sections define the ingredients and the physical import gate. We now make explicit what it means for a calculation, a code path, or an extraction to carry the label $\text{N}^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$. The goal is to make the eventual paper’s claim precise: the paper may present a generic $\text{N}^3\text{LL}'$ prescription in k_T space, while any numerical implementation or data extraction must satisfy the concrete certificate below.

23.1 Prescription label versus instantiated-implementation label

There are two logically distinct labels.

1. **Formal prescription label.** A paper-level formula may be called an $N^3LL'_{\mathcal{K}_{\text{CSS2}}}$ prescription if every perturbative object is specified by the same ingredient content as the parent CSS2 calculation, and if all b_T logarithms are mapped analytically to k_T -space distributions by the rules of Sections 21 and 22. This is a statement about the perturbative construction, not about any particular nonperturbative model.
2. **Instantiated implementation label.** A concrete code run or extracted TMD may advertise an $N^3LL'_{\mathcal{K}_{\text{CSS2}}}$ perturbative label only after the physical coefficient payload has been staged, hashed, converted to the canonical schema, and validated through the $\mathcal{O}(a_s^3)$ expansion gate.

Thus the label is not attached to the DNN, Gaussian ansatz, spline basis, or other nonperturbative module. It is attached to the perturbative W operator:

$$W^{\mathcal{K}_{\text{CSS2}}, N^3LL'} = H^{[3]} \otimes \mathcal{U}^{N^3LL} \otimes \mathcal{C}^{[3]} \otimes f, \quad (304)$$

where the transverse parts of the products are convolution products in the $\{\delta^{(2)}, \mathcal{L}_n\}$ distribution algebra. The nonperturbative module is appended only after this operator has been defined:

$$F_{i/h}^{\text{full}} = \mathcal{P}_{i/h}^{N^3LL'}[f_h; H, D, \mathcal{C}, P, \beta, \Gamma, \gamma_F] \otimes_T \mathcal{M}_{i/h}^{\text{NP}} + \text{overlap or power corrections}. \quad (305)$$

This is the intended publication-level separation: the perturbative accuracy is universal, while the soft/nonperturbative choice is modular.

23.2 Minimal N3LL-prime certificate

Define a certificate

$$\mathfrak{C}_{N^3LL'} = \{\text{scheme, ingredients, coefficient provenance, basis, RG, expansion, process, NP separation}\}. \quad (306)$$

The certificate is accepted only if

$$\begin{aligned} \text{ClaimAllowed}_{N^3LL'} &= \text{ParentScheme}_{\mathcal{K}_{\text{CSS2}}} \wedge \text{IngredientComplete}_{N^3LL'} \wedge \text{PhysicalPayload} \\ &\wedge \text{NoMockCoefficients} \wedge \text{NoNumericalBTransform} \wedge \text{RGConsistent} \\ &\wedge \text{ExpandedTo}\{a_s^3\} \wedge \text{NPSeparated}. \end{aligned} \quad (307)$$

The ingredients required by $\text{IngredientComplete}_{N^3LL'}$ are the manifest of Section 20: four-loop cusp anomalous dimension, four-loop QCD beta function, three-loop noncusp TMD anomalous dimension, three-loop Collins-Soper/rapidity kernel, three-loop hard function, three-loop TMD matching coefficients, and DGLAP splitting functions through $P^{(2)}$. For full-spectrum predictions outside the singular region, the separate Y -term also requires the chosen fixed-order accuracy.

The implementation-level certificate has two modes:

Mode	What may be claimed	What may not be claimed
Formal prescription	A generic pure- k_T N ³ LL' construction equivalent to CSS2 by ingredient map and distribution algebra.	A completed numerical extraction or a verified physical coefficient import.
Implementation dry run	Importer, generator algebra, RG, and expansion machinery using mock or partial payloads.	A physical N ³ LL' prediction.
Certified implementation	A physical N ³ LL' _{KCSS2} perturbative W operator for the declared process class.	Universality beyond the declared process, flavor basis, or nonperturbative module assumptions.

Working decision. The paper can legitimately present an N³LL' prescription once the ingredient map, distribution algebra, and generic operator are stated. A numerical extraction can only carry the N³LL' label after Eq. (307) evaluates to true with physical coefficient payloads.

23.3 Physical payloads required to unlock the label

For a Drell–Yan TMDPDF-only implementation, the minimal physical payload is

$$\mathcal{P}_{\text{DY}}^{\text{min}} = \left\{ H_{q\bar{q}}^{(0,1,2,3)}, C_{i \leftarrow j}^{(0,1,2,3)}, D_i^{(1,2,3)}, \Gamma_i^{(0,1,2,3)}, \gamma_{F,i}^{(0,1,2)}, \beta_{0,1,2,3}, P_{ij}^{(0,1,2)} \right\}. \quad (308)$$

For SIDIS or back-to-back hadron production in e^+e^- , the payload must additionally include TMDFF matching through three loops and the appropriate timelike DGLAP kernels. For gluon-initiated color-singlet processes, the gluon hard function and gluon matching channels must pass the same source and expansion checks.

The public high-order source records justify the availability of these matching inputs: EMV2020 states that the N³LO quark and gluon TMDPDF matching kernels are the last missing ingredient for N³LL' q_T spectra, and LYZZ2021 provides N³LO unpolarized quark/gluon TMDPDF and TMDFF matching coefficients with arXiv ancillary files [17, 19]. The present note does not claim that those files have been parsed unless their hashes appear in the local coefficient ledger.

23.4 Coefficient staging kit

Since the sandbox used for this note does not have live network access from the code runtime, this note includes a reproducible staging kit rather than pretending to have imported the physical files. The script

```
kt_kcss_stage_public_coefficients.sh
```

records the official arXiv source packages and ancillary-file paths for EMV2020 and LYZZ2021. When run in an environment with network access, it should populate

```
/mnt/data/kcss_coeff_sources/2006.05329v2/
/mnt/data/kcss_coeff_sources/2012.03256v2/
```

then compute SHA256 hashes. The certificate validator refuses to accept files without these hashes and refuses to accept mock coefficients for physics mode.

23.5 Current certificate status

In the present run, the formal-prescription part of the certificate is active, while the physical implementation part remains blocked by the absence of local source-stamped coefficient files:

Formal $N^3LL'_{\mathcal{K}_{\text{CSS2}}}$ prescription: **specified**. Mock coefficient payload: **forbidden for physics**. Physical coefficient import: **not yet staged in this sandbox**. Implementation-level N^3LL' claim: **blocked until source hashes and $\mathcal{O}(a_s^3)$ expansion gates pass**.

This is the disciplined route to the label: the paper can present the pure- k_T N^3LL' prescription generically, and the first phenomenological implementation earns the label by completing the certificate.

24 Full-spectrum matching and perturbative uncertainties

The singular operator $W^{\mathcal{K}_{\text{CSS2}}}$ is the object that carries the TMD resummation accuracy. For a full transverse-momentum spectrum it must be combined with a nonsingular fixed-order remainder,

$$d\sigma = d\sigma_W^{\mathcal{K}_{\text{CSS2}}, N^k LL'} + d\sigma_Y + d\sigma_{\text{pow}}, \quad (309)$$

with

$$d\sigma_Y = d\sigma_{\text{FO}}^{[m]} - \left[d\sigma_W^{\mathcal{K}_{\text{CSS2}}, N^k LL'} \right]_{\text{expanded to } \mathcal{O}(a_s^m)}. \quad (310)$$

The subtraction in Eq. (310) must use the same PDFs, hard normalization, flavor basis, scale choices, electroweak inputs, and bin definitions as the fixed-order calculation. External programs such as DYTurbo, SCETlib, CuTe-MCFM/MCFM, or other fixed-order/resummed implementations are benchmark and provider tools for $d\sigma_{\text{FO}}$ and matched spectra [37, 38, 39, 34]; they do not define the universal TMD legs.

The two tails remain distinct:

$$\boxed{\text{TMD large-}k_T \text{ tail}} \iff \mathcal{C}_{i \leftarrow j} \otimes_x f_j, \quad \boxed{\text{cross-section large-}q_T \text{ tail}} \iff Y. \quad (311)$$

Conflating these two objects would turn a universal TMD extraction into a process-specific spectrum fit.

For the formalism paper the perturbative uncertainty budget is separated from the nonperturbative model uncertainty:

$$\Delta_{\text{pert}} = \Delta_{\text{scale}} \oplus \Delta_{\text{match}} \oplus \Delta_{\text{scheme}} \oplus \Delta_{\text{numerical}}. \quad (312)$$

Here Δ_{scale} comes from variations of μ_H , boundary scales, final scales, and rapidity scales subject to constraints such as $\zeta_A \zeta_B = Q^4$ in Drell-Yan; Δ_{match} comes from the W - Y transition and fixed-order truncation; Δ_{scheme} comes from any finite conversion between $\mathcal{K}_{\text{CSS2}}$ and a future child scheme; and $\Delta_{\text{numerical}}$ comes from binned distribution evaluation, quadrature, interpolation, and convolution-grid errors. Experimental errors, PDF-replica errors, and nonperturbative model/DNN uncertainties belong to the later extraction paper.

25 Paper-ready claim table and extraction workflow

The first paper should make a formalism claim, not a completed data-fit claim. The appropriate claim hierarchy is:

Claim	Allowed when	Not implied
Formal $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ prescription	The ingredient map, distribution basis, convolution algebra, evolution kernel, matching import contract, and $W + Y$ subtraction formula are specified.	A numerical extraction has been performed.
Numerical $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ implementation	Physical three-loop coefficient payloads are staged, hashed, parsed, converted, and the $\mathcal{O}(a_s^3)$ singular gate passes.	Data have been fitted.
k_T -space $N^3\text{LL}'$ extraction	The numerical operator is combined with a specified nonperturbative module, data set, fit protocol, and uncertainty model.	The nonperturbative module is unique.
DNN-based extraction	The chosen $\mathcal{M}_{\text{NP}}$ is a neural parameterization satisfying the admissibility conditions of Sec. 18.	The perturbative label is produced by the DNN.

A paper based on this note can therefore be organized as follows.

1. Motivation: why a direct- k_T extraction is useful and what it does not claim to invent.
2. Parent scheme: definition of $\mathcal{K}_{\text{CSS2}}$ as a k_T -space representation of CSS2.
3. Distribution algebra: plus distributions, generators, bin actions, and convolution exponentials.
4. Perturbative evolution and matching: D_i , $\mathcal{C}_{i \leftarrow j}$, hard functions, DGLAP, and the $N^3\text{LL}'$ ingredient map.
5. Process-level W operator: DY, SIDIS, e^+e^- , and gluon-channel templates.
6. Nonperturbative module: admissibility conditions and examples, including but not limited to DNNs.
7. Regulated normalization and cumulants.
8. $W + Y$ matching and perturbative uncertainties.
9. Validation hierarchy and implementation certificate.
10. Outlook toward a genuine numerical $N^3\text{LL}'$ extraction.

26 Numerical representation

Because perturbative kernels are distributions, the numerical implementation should avoid unbinned pointwise evaluation near $q_T = 0$. Preferred representations are:

1. bin-integrated kernels matching experimental binning;
2. cumulants $\Sigma(q_T^{\max}) = \int^{q_T^{\max}} d^2\mathbf{q} W(\mathbf{q})$;
3. smooth test-function projections;
4. radial basis expansions with analytically known integrals over bins.

The transverse convolution of radial functions can be evaluated through the angular integral

$$(A \otimes_T B)(q) = \int_0^\infty dk k A(k) \int_0^{2\pi} d\phi B\left(\sqrt{q^2 + k^2 - 2qk \cos \phi}\right), \quad (313)$$

for ordinary functions. Distributional pieces should be handled analytically before numerical quadrature.

Working decision. The paper-level recommendation is the hybrid approach: singular kernels are handled analytically through bin or test-function actions, while ordinary nonperturbative modules are represented on radial grids or flexible bases and combined with analytic subtraction terms.

27 Validation sequence

The formal prescription becomes a numerical implementation only after a validation ladder has been completed. The required checks are:

1. **Distribution algebra.** Validate $\delta^{(2)}$, $\mathcal{L}_n$, $\mathfrak{L}_n$, generator derivatives, bin actions, and transverse convolution identities.
2. **One-loop singular spectrum.** Reproduce the $A_q^{(1)} = C_F$ and $B_q^{(1)} = -3C_F/2$ Drell-Yan singular spectrum in bins, including the correct $d^2\mathbf{q}$ versus dq_T^2 normalization.
3. **Flavor-vector and x -convolution tests.** Validate the longitudinal plus-prescription, DGLAP kernels, and quark/gluon mixing independently of the transverse plus prescription.
4. **Evolution tests.** Check path independence in (μ, ζ) , semigroup closure, fixed-coupling limits, and running-coupling endpoint identities.
5. **NNLO singular gate.** Import the physical $\mathcal{C}^{(2)}$, $H^{(2)}$, and $P^{(1)}$ ingredients; expand $W^{\mathcal{K}_{\text{CSS2}}}$ through $\mathcal{O}(a_s^2)$ and compare bin by bin with the parent CSS2 singular expansion.
6. **N³LO singular gate.** Import the physical $\mathcal{C}^{(3)}$, $H^{(3)}$, $D^{(3)}$, and $P^{(2)}$ ingredients; expand through $\mathcal{O}(a_s^3)$ and compare bin by bin with the parent CSS2 singular expansion.
7. **Y -term consistency.** Add Eq. (310); verify that the matched result tends to the resummed result at small q_T and to fixed order at large q_T .
8. **Nonperturbative closure.** Generate pseudo-data from a known $\mathcal{M}_{\text{NP}}$ and recover it without allowing the model to absorb missing perturbative ingredients.

Only after this ladder is passed should a data fit advertise an implementation-level $N^3\text{LL}'$ extraction.

28 Relation to Fernando-Keller pipeline

Fernando-Keller can be summarized schematically as

$$\text{data} \longrightarrow \text{learned structure kernel} \longrightarrow s(x, k; Q) \longrightarrow F_{q/h}(x, k; Q) \simeq f_{q/h}(x, Q) s(x, k; Q). \quad (314)$$

The perturbative upgrade is instead

$$\text{data} \longrightarrow N^k \text{LL}_{k_T} \text{ forward operator} \longrightarrow F_{i/h}^{\text{core}}(x, k; \theta) \longrightarrow F_{i/h}^{\text{pert+NP-sub}}(x, k; \mu, \zeta). \quad (315)$$

The neural network is moved from “the full transverse profile” to “the nonperturbative boundary condition.”

29 Closure of the formalism note

The formalism note is now complete for the purpose of drafting the first paper. It has resolved the conceptual questions as follows.

1. **Scheme.** The parent construction is $\mathcal{K}_{\text{CSS2}}$, a direct k_T -space representation of Collins/CSS2. Future child schemes require explicit finite conversion kernels and inverse kernels.
2. **Distribution basis.** The conventional $\mathcal{L}_n$ basis is used for bin actions and comparison to singular spectra; the generator basis is used for convolution algebra and Sudakov exponentiation.
3. **Perturbative tail.** The universal TMD large- k_T tail is fixed by $\mathcal{C}_{i \leftarrow j} \otimes_x f_j$. The large- q_T cross-section tail is the Y term. These are distinct objects.
4. **Normalization.** Physical constraints are regulated cumulants and weighted moments, not unregulated full- k_T integrals.
5. **Uncertainty.** The formalism note defines perturbative uncertainty classes only. Experimental, PDF-replica, and nonperturbative-model uncertainties are extraction-level additions.
6. **Flavor.** The formalism is flavor-vector valued from the start. Any simplified flavor model is a data-fit choice, not a property of $\mathcal{K}_{\text{CSS2}}$.
7. **DNN role.** A DNN is one possible realization of $\mathcal{M}_{\text{NP}}$. It is not the source of the perturbative accuracy label.

The remaining tasks belong to the implementation and phenomenology program, not to the formal definition. They are: stage and hash the physical NNLO/N³LO coefficient files; convert them to the canonical CSS2-compatible schema; complete the $\mathcal{O}(a_s^2)$ and $\mathcal{O}(a_s^3)$ singular gates; benchmark $W + Y$ against external Drell-Yan codes; choose a specific $\mathcal{M}_{\text{NP}}$; and perform closure tests and data fits. Those tasks are the basis for the later numerical extraction paper.

A Minimal notation dictionary

Symbol	Meaning
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$F_{i/h}(x, \mathbf{k}; \mu, \zeta)$	Renormalized TMDPDF for parton i in hadron h in transverse-momentum space.
$H_{ij}(Q, \mu)$	Hard function for the partonic channel ij .
$\mathcal{D}_i(\mathbf{k}; \mu)$	Momentum-space Collins-Soper kernel.
$\mathcal{U}_i$	Evolution Green's function acting by transverse convolution.
$\mathcal{C}_{i \leftarrow j}$	Perturbative matching coefficient from collinear PDF j to TMD parton i .
W	Resummed/singular TMD contribution to the cross section.
Y	Fixed-order remainder: full fixed order minus the fixed-order expansion of W .
F^{core}	Neural or parametric nonperturbative core at the boundary scale.
F^{sub}	Overlap subtraction between the perturbative tail and nonperturbative core.

B Reference anchors

- Foundations of the baseline b_T framework and scheme translation: Collins-Soper, CSS, Collins 2011, Collins-Rogers, and the TMD Handbook [1, 2, 3, 36, 4].
- Early direct- q_T and transverse-momentum-space resummation: Ellis-Veseli, Frixione-Nason-Ridolfi, and Kulesza-Stirling [5, 6, 7, 8].
- Modern direct-space and distribution-space resummation: Monni-Re-Torrielli, Bizon et al., Ebert-Tackmann, and Simonelli [9, 10, 11, 12].
- Perturbative ingredients for a high-order k_T implementation: RRG soft/beam functions, $N^3\text{LO}$ TMD matching, high-order rapidity anomalous dimensions, four-loop cusp information, three-loop hard functions, and modern $N^3\text{LL}'$ Drell-Yan benchmarks [16, 17, 19, 15, 20, 21, 30, 31, 34].
- Integral constraints and large- k_T consistency: Ebert-Michel-Stewart-Sun, Gonzalez-Hernandez-Rainaldi-Rogers, and HSO phenomenology [22, 23, 24].
- Existing high-accuracy and neural TMD extractions: Moos-Scimemi-Vladimirov-Zurita and MAP neural-network fits [25, 26, 27].
- Direct k_T inverse reconstruction template: Fernando-Keller [35].
- Cross-section fixed-order and matched benchmarks for the Y term: DYTurbo and CuTe-MCFM/MCFM [37, 38, 39].
- Momentum-space TMD evolution in a neighboring framework: Parton Branching and PB-to-CSS comparisons [29, 28].

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